Chapter 03 | Functions of a real variable

Worked solutions

Original, unofficial exercises. Source pages guide conceptual coverage only. Always check the domain assumptions.

L37 | Functions, domains and graphs

Source: PDF pages 2, 3, 4

1/5 | Apprentice

L37-P001 | Let f(x) = 2x - 4. Find f(-3).

f(-3) = 2×(-3) - 4 = -10.

L37-P002 | Let f(x) = 3x - 3. Find f(-1).

f(-1) = 3×(-1) - 3 = -6.

L37-P003 | Let f(x) = 4x - 2. Find f(1).

f(1) = 4×(1) - 2 = 2.

L37-P004 | Let f(x) = 5x - 1. Find f(3).

f(3) = 5×(3) - 1 = 14.

L37-P005 | Let f(x) = 6x + 0. Find f(-2).

f(-2) = 6×(-2) + 0 = -12.

L37-P006 | Let f(x) = 2x + 1. Find f(0).

f(0) = 2×(0) + 1 = 1.

L37-P007 | Let f(x) = 3x + 2. Find f(2).

f(2) = 3×(2) + 2 = 8.

L37-P008 | Let f(x) = 4x + 3. Find f(-3).

f(-3) = 4×(-3) + 3 = -9.

L37-P009 | Let f(x) = 5x + 4. Find f(-1).

f(-1) = 5×(-1) + 4 = -1.

L37-P010 | Let f(x) = 6x - 4. Find f(1).

f(1) = 6×(1) - 4 = 2.

L37-P011 | Let f(x) = 2x - 3. Find f(3).

f(3) = 2×(3) - 3 = 3.

L37-P012 | Let f(x) = 3x - 2. Find f(-2).

f(-2) = 3×(-2) - 2 = -8.

L37-P013 | Let f(x) = 4x - 1. Find f(0).

f(0) = 4×(0) - 1 = -1.

L37-P014 | Let f(x) = 5x + 0. Find f(2).

f(2) = 5×(2) + 0 = 10.

L37-P015 | Let f(x) = 6x + 1. Find f(-3).

f(-3) = 6×(-3) + 1 = -17.

L37-P016 | Let f(x) = 2x + 2. Find f(-1).

f(-1) = 2×(-1) + 2 = 0.

L37-P017 | Let f(x) = 3x + 3. Find f(1).

f(1) = 3×(1) + 3 = 6.

L37-P018 | Let f(x) = 4x + 4. Find f(3).

f(3) = 4×(3) + 4 = 16.

L37-P019 | Let f(x) = 5x - 4. Find f(-2).

f(-2) = 5×(-2) - 4 = -14.

L37-P020 | Let f(x) = 6x - 3. Find f(0).

f(0) = 6×(0) - 3 = -3.

2/5 | Practitioner

L37-P021 | f(x) = 1x^2 - 4x + 1. Evaluate f(-3).

Substitute x = -3: 1×(-3)^2 - 4×(-3) + 1 = 22.

L37-P022 | f(x) = 2x^2 - 3x + 2. Evaluate f(-1).

Substitute x = -1: 2×(-1)^2 - 3×(-1) + 2 = 7.

L37-P023 | f(x) = 3x^2 - 2x + 3. Evaluate f(1).

Substitute x = 1: 3×(1)^2 - 2×(1) + 3 = 4.

L37-P024 | f(x) = 1x^2 - 1x + 4. Evaluate f(3).

Substitute x = 3: 1×(3)^2 - 1×(3) + 4 = 10.

L37-P025 | f(x) = 2x^2 + 0x + 1. Evaluate f(-2).

Substitute x = -2: 2×(-2)^2 + 0×(-2) + 1 = 9.

L37-P026 | f(x) = 3x^2 + 1x + 2. Evaluate f(0).

Substitute x = 0: 3×(0)^2 + 1×(0) + 2 = 2.

L37-P027 | f(x) = 1x^2 + 2x + 3. Evaluate f(2).

Substitute x = 2: 1×(2)^2 + 2×(2) + 3 = 11.

L37-P028 | f(x) = 2x^2 + 3x + 4. Evaluate f(-3).

Substitute x = -3: 2×(-3)^2 + 3×(-3) + 4 = 13.

L37-P029 | f(x) = 3x^2 + 4x + 1. Evaluate f(-1).

Substitute x = -1: 3×(-1)^2 + 4×(-1) + 1 = 0.

L37-P030 | f(x) = 1x^2 - 4x + 2. Evaluate f(1).

Substitute x = 1: 1×(1)^2 - 4×(1) + 2 = -1.

L37-P031 | f(x) = 2x^2 - 3x + 3. Evaluate f(3).

Substitute x = 3: 2×(3)^2 - 3×(3) + 3 = 12.

L37-P032 | f(x) = 3x^2 - 2x + 4. Evaluate f(-2).

Substitute x = -2: 3×(-2)^2 - 2×(-2) + 4 = 20.

L37-P033 | f(x) = 1x^2 - 1x + 1. Evaluate f(0).

Substitute x = 0: 1×(0)^2 - 1×(0) + 1 = 1.

L37-P034 | f(x) = 2x^2 + 0x + 2. Evaluate f(2).

Substitute x = 2: 2×(2)^2 + 0×(2) + 2 = 10.

L37-P035 | f(x) = 3x^2 + 1x + 3. Evaluate f(-3).

Substitute x = -3: 3×(-3)^2 + 1×(-3) + 3 = 27.

L37-P036 | f(x) = 1x^2 + 2x + 4. Evaluate f(-1).

Substitute x = -1: 1×(-1)^2 + 2×(-1) + 4 = 3.

L37-P037 | f(x) = 2x^2 + 3x + 1. Evaluate f(1).

Substitute x = 1: 2×(1)^2 + 3×(1) + 1 = 6.

L37-P038 | f(x) = 3x^2 + 4x + 2. Evaluate f(3).

Substitute x = 3: 3×(3)^2 + 4×(3) + 2 = 41.

L37-P039 | f(x) = 1x^2 - 4x + 3. Evaluate f(-2).

Substitute x = -2: 1×(-2)^2 - 4×(-2) + 3 = 15.

L37-P040 | f(x) = 2x^2 - 3x + 4. Evaluate f(0).

Substitute x = 0: 2×(0)^2 - 3×(0) + 4 = 4.

3/5 | Challenger

L37-P041 | Find the domain of f(x) = (x^2 + 1)/((x - 2)(x + 1)).

The denominator is zero for x = 2 or x = -1. Domain: all real numbers except those values.

L37-P042 | Find the domain of f(x) = (x^2 + 2)/((x - 3)(x + 2)).

The denominator is zero for x = 3 or x = -2. Domain: all real numbers except those values.

L37-P043 | Find the domain of f(x) = (x^2 + 3)/((x - 4)(x + 3)).

The denominator is zero for x = 4 or x = -3. Domain: all real numbers except those values.

L37-P044 | Find the domain of f(x) = (x^2 + 4)/((x - 5)(x + 4)).

The denominator is zero for x = 5 or x = -4. Domain: all real numbers except those values.

L37-P045 | Find the domain of f(x) = (x^2 + 1)/((x - 6)(x + 1)).

The denominator is zero for x = 6 or x = -1. Domain: all real numbers except those values.

L37-P046 | Find the domain of f(x) = (x^2 + 2)/((x - 2)(x + 2)).

The denominator is zero for x = 2 or x = -2. Domain: all real numbers except those values.

L37-P047 | Find the domain of f(x) = (x^2 + 3)/((x - 3)(x + 3)).

The denominator is zero for x = 3 or x = -3. Domain: all real numbers except those values.

L37-P048 | Find the domain of f(x) = (x^2 + 4)/((x - 4)(x + 4)).

The denominator is zero for x = 4 or x = -4. Domain: all real numbers except those values.

L37-P049 | Find the domain of f(x) = (x^2 + 1)/((x - 5)(x + 1)).

The denominator is zero for x = 5 or x = -1. Domain: all real numbers except those values.

L37-P050 | Find the domain of f(x) = (x^2 + 2)/((x - 6)(x + 2)).

The denominator is zero for x = 6 or x = -2. Domain: all real numbers except those values.

L37-P051 | Find the domain of f(x) = (x^2 + 3)/((x - 2)(x + 3)).

The denominator is zero for x = 2 or x = -3. Domain: all real numbers except those values.

L37-P052 | Find the domain of f(x) = (x^2 + 4)/((x - 3)(x + 4)).

The denominator is zero for x = 3 or x = -4. Domain: all real numbers except those values.

L37-P053 | Find the domain of f(x) = (x^2 + 1)/((x - 4)(x + 1)).

The denominator is zero for x = 4 or x = -1. Domain: all real numbers except those values.

L37-P054 | Find the domain of f(x) = (x^2 + 2)/((x - 5)(x + 2)).

The denominator is zero for x = 5 or x = -2. Domain: all real numbers except those values.

L37-P055 | Find the domain of f(x) = (x^2 + 3)/((x - 6)(x + 3)).

The denominator is zero for x = 6 or x = -3. Domain: all real numbers except those values.

L37-P056 | Find the domain of f(x) = (x^2 + 4)/((x - 2)(x + 4)).

The denominator is zero for x = 2 or x = -4. Domain: all real numbers except those values.

L37-P057 | Find the domain of f(x) = (x^2 + 1)/((x - 3)(x + 1)).

The denominator is zero for x = 3 or x = -1. Domain: all real numbers except those values.

L37-P058 | Find the domain of f(x) = (x^2 + 2)/((x - 4)(x + 2)).

The denominator is zero for x = 4 or x = -2. Domain: all real numbers except those values.

L37-P059 | Find the domain of f(x) = (x^2 + 3)/((x - 5)(x + 3)).

The denominator is zero for x = 5 or x = -3. Domain: all real numbers except those values.

L37-P060 | Find the domain of f(x) = (x^2 + 4)/((x - 6)(x + 4)).

The denominator is zero for x = 6 or x = -4. Domain: all real numbers except those values.

4/5 | Expert

L37-P061 | f(x) = 2x - 4. Solve f(x) = -10 and state the corresponding graph point.

2x - 4 = -10, so 2x = -6 and x = -3. Point: (-3, -10).

L37-P062 | f(x) = 3x - 3. Solve f(x) = -6 and state the corresponding graph point.

3x - 3 = -6, so 3x = -3 and x = -1. Point: (-1, -6).

L37-P063 | f(x) = 4x - 2. Solve f(x) = 2 and state the corresponding graph point.

4x - 2 = 2, so 4x = 4 and x = 1. Point: (1, 2).

L37-P064 | f(x) = 5x - 1. Solve f(x) = 14 and state the corresponding graph point.

5x - 1 = 14, so 5x = 15 and x = 3. Point: (3, 14).

L37-P065 | f(x) = 6x + 0. Solve f(x) = -12 and state the corresponding graph point.

6x + 0 = -12, so 6x = -12 and x = -2. Point: (-2, -12).

L37-P066 | f(x) = 2x + 1. Solve f(x) = 1 and state the corresponding graph point.

2x + 1 = 1, so 2x = 0 and x = 0. Point: (0, 1).

L37-P067 | f(x) = 3x + 2. Solve f(x) = 8 and state the corresponding graph point.

3x + 2 = 8, so 3x = 6 and x = 2. Point: (2, 8).

L37-P068 | f(x) = 4x + 3. Solve f(x) = -9 and state the corresponding graph point.

4x + 3 = -9, so 4x = -12 and x = -3. Point: (-3, -9).

L37-P069 | f(x) = 5x + 4. Solve f(x) = -1 and state the corresponding graph point.

5x + 4 = -1, so 5x = -5 and x = -1. Point: (-1, -1).

L37-P070 | f(x) = 6x - 4. Solve f(x) = 2 and state the corresponding graph point.

6x - 4 = 2, so 6x = 6 and x = 1. Point: (1, 2).

L37-P071 | f(x) = 2x - 3. Solve f(x) = 3 and state the corresponding graph point.

2x - 3 = 3, so 2x = 6 and x = 3. Point: (3, 3).

L37-P072 | f(x) = 3x - 2. Solve f(x) = -8 and state the corresponding graph point.

3x - 2 = -8, so 3x = -6 and x = -2. Point: (-2, -8).

L37-P073 | f(x) = 4x - 1. Solve f(x) = -1 and state the corresponding graph point.

4x - 1 = -1, so 4x = 0 and x = 0. Point: (0, -1).

L37-P074 | f(x) = 5x + 0. Solve f(x) = 10 and state the corresponding graph point.

5x + 0 = 10, so 5x = 10 and x = 2. Point: (2, 10).

L37-P075 | f(x) = 6x + 1. Solve f(x) = -17 and state the corresponding graph point.

6x + 1 = -17, so 6x = -18 and x = -3. Point: (-3, -17).

L37-P076 | f(x) = 2x + 2. Solve f(x) = 0 and state the corresponding graph point.

2x + 2 = 0, so 2x = -2 and x = -1. Point: (-1, 0).

L37-P077 | f(x) = 3x + 3. Solve f(x) = 6 and state the corresponding graph point.

3x + 3 = 6, so 3x = 3 and x = 1. Point: (1, 6).

L37-P078 | f(x) = 4x + 4. Solve f(x) = 16 and state the corresponding graph point.

4x + 4 = 16, so 4x = 12 and x = 3. Point: (3, 16).

L37-P079 | f(x) = 5x - 4. Solve f(x) = -14 and state the corresponding graph point.

5x - 4 = -14, so 5x = -10 and x = -2. Point: (-2, -14).

L37-P080 | f(x) = 6x - 3. Solve f(x) = -3 and state the corresponding graph point.

6x - 3 = -3, so 6x = 0 and x = 0. Point: (0, -3).

5/5 | Ninja

L37-P081 | Let g(x) = 2x + 1 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

Require 2x + 1 ≠ 2, hence x ≠ 1/2. At x = 1, g(x) = 3, giving f(g(x)) = 1.

L37-P082 | Let g(x) = 3x + 2 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

Require 3x + 2 ≠ 3, hence x ≠ 1/3. At x = 2, g(x) = 8, giving f(g(x)) = 1/5.

L37-P083 | Let g(x) = 4x + 3 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

Require 4x + 3 ≠ 4, hence x ≠ 1/4. At x = 3, g(x) = 15, giving f(g(x)) = 1/11.

L37-P084 | Let g(x) = 5x + 1 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

Require 5x + 1 ≠ 5, hence x ≠ 4/5. At x = 4, g(x) = 21, giving f(g(x)) = 1/16.

L37-P085 | Let g(x) = 2x + 2 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

Require 2x + 2 ≠ 2, hence x ≠ 0. At x = 1, g(x) = 4, giving f(g(x)) = 1/2.

L37-P086 | Let g(x) = 3x + 3 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

Require 3x + 3 ≠ 3, hence x ≠ 0. At x = 2, g(x) = 9, giving f(g(x)) = 1/6.

L37-P087 | Let g(x) = 4x + 1 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

Require 4x + 1 ≠ 4, hence x ≠ 3/4. At x = 3, g(x) = 13, giving f(g(x)) = 1/9.

L37-P088 | Let g(x) = 5x + 2 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

Require 5x + 2 ≠ 5, hence x ≠ 3/5. At x = 4, g(x) = 22, giving f(g(x)) = 1/17.

L37-P089 | Let g(x) = 2x + 3 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

Require 2x + 3 ≠ 2, hence x ≠ -1/2. At x = 1, g(x) = 5, giving f(g(x)) = 1/3.

L37-P090 | Let g(x) = 3x + 1 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

Require 3x + 1 ≠ 3, hence x ≠ 2/3. At x = 2, g(x) = 7, giving f(g(x)) = 1/4.

L37-P091 | Let g(x) = 4x + 2 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

Require 4x + 2 ≠ 4, hence x ≠ 1/2. At x = 3, g(x) = 14, giving f(g(x)) = 1/10.

L37-P092 | Let g(x) = 5x + 3 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

Require 5x + 3 ≠ 5, hence x ≠ 2/5. At x = 4, g(x) = 23, giving f(g(x)) = 1/18.

L37-P093 | Let g(x) = 2x + 1 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

Require 2x + 1 ≠ 2, hence x ≠ 1/2. At x = 1, g(x) = 3, giving f(g(x)) = 1.

L37-P094 | Let g(x) = 3x + 2 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

Require 3x + 2 ≠ 3, hence x ≠ 1/3. At x = 2, g(x) = 8, giving f(g(x)) = 1/5.

L37-P095 | Let g(x) = 4x + 3 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

Require 4x + 3 ≠ 4, hence x ≠ 1/4. At x = 3, g(x) = 15, giving f(g(x)) = 1/11.

L37-P096 | Let g(x) = 5x + 1 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

Require 5x + 1 ≠ 5, hence x ≠ 4/5. At x = 4, g(x) = 21, giving f(g(x)) = 1/16.

L37-P097 | Let g(x) = 2x + 2 and f(u) = 1/(u - 2). Find the domain of f∘g, then (f∘g)(1).

Require 2x + 2 ≠ 2, hence x ≠ 0. At x = 1, g(x) = 4, giving f(g(x)) = 1/2.

L37-P098 | Let g(x) = 3x + 3 and f(u) = 1/(u - 3). Find the domain of f∘g, then (f∘g)(2).

Require 3x + 3 ≠ 3, hence x ≠ 0. At x = 2, g(x) = 9, giving f(g(x)) = 1/6.

L37-P099 | Let g(x) = 4x + 1 and f(u) = 1/(u - 4). Find the domain of f∘g, then (f∘g)(3).

Require 4x + 1 ≠ 4, hence x ≠ 3/4. At x = 3, g(x) = 13, giving f(g(x)) = 1/9.

L37-P100 | Let g(x) = 5x + 2 and f(u) = 1/(u - 5). Find the domain of f∘g, then (f∘g)(4).

Require 5x + 2 ≠ 5, hence x ≠ 3/5. At x = 4, g(x) = 22, giving f(g(x)) = 1/17.

L38 | Average rates, derivatives and tangents

Source: PDF pages 5, 6, 7, 8, 9, 10, 11, 12

1/5 | Apprentice

L38-P001 | For f(t) = 1x - 3, find (f(-1) - f(-2))/1.

An affine function has a constant average rate: (1×1)/1 = 1.

L38-P002 | For f(t) = 2x - 2, find (f(1) - f(-1))/2.

An affine function has a constant average rate: (2×2)/2 = 2.

L38-P003 | For f(t) = 3x - 1, find (f(3) - f(0))/3.

An affine function has a constant average rate: (3×3)/3 = 3.

L38-P004 | For f(t) = 4x + 0, find (f(2) - f(1))/1.

An affine function has a constant average rate: (4×1)/1 = 4.

L38-P005 | For f(t) = 1x + 1, find (f(4) - f(2))/2.

An affine function has a constant average rate: (1×2)/2 = 1.

L38-P006 | For f(t) = 2x + 2, find (f(1) - f(-2))/3.

An affine function has a constant average rate: (2×3)/3 = 2.

L38-P007 | For f(t) = 3x + 3, find (f(0) - f(-1))/1.

An affine function has a constant average rate: (3×1)/1 = 3.

L38-P008 | For f(t) = 4x - 3, find (f(2) - f(0))/2.

An affine function has a constant average rate: (4×2)/2 = 4.

L38-P009 | For f(t) = 1x - 2, find (f(4) - f(1))/3.

An affine function has a constant average rate: (1×3)/3 = 1.

L38-P010 | For f(t) = 2x - 1, find (f(3) - f(2))/1.

An affine function has a constant average rate: (2×1)/1 = 2.

L38-P011 | For f(t) = 3x + 0, find (f(0) - f(-2))/2.

An affine function has a constant average rate: (3×2)/2 = 3.

L38-P012 | For f(t) = 4x + 1, find (f(2) - f(-1))/3.

An affine function has a constant average rate: (4×3)/3 = 4.

L38-P013 | For f(t) = 1x + 2, find (f(1) - f(0))/1.

An affine function has a constant average rate: (1×1)/1 = 1.

L38-P014 | For f(t) = 2x + 3, find (f(3) - f(1))/2.

An affine function has a constant average rate: (2×2)/2 = 2.

L38-P015 | For f(t) = 3x - 3, find (f(5) - f(2))/3.

An affine function has a constant average rate: (3×3)/3 = 3.

L38-P016 | For f(t) = 4x - 2, find (f(-1) - f(-2))/1.

An affine function has a constant average rate: (4×1)/1 = 4.

L38-P017 | For f(t) = 1x - 1, find (f(1) - f(-1))/2.

An affine function has a constant average rate: (1×2)/2 = 1.

L38-P018 | For f(t) = 2x + 0, find (f(3) - f(0))/3.

An affine function has a constant average rate: (2×3)/3 = 2.

L38-P019 | For f(t) = 3x + 1, find (f(2) - f(1))/1.

An affine function has a constant average rate: (3×1)/1 = 3.

L38-P020 | For f(t) = 4x + 2, find (f(4) - f(2))/2.

An affine function has a constant average rate: (4×2)/2 = 4.

2/5 | Practitioner

L38-P021 | For f(x) = 1x^2, compute the secant slope from x = -2 to x = -1.

[f(p+h)-f(p)]/h = 1[(p+h)^2-p^2]/h = 1(2p+h) = -3.

L38-P022 | For f(x) = 2x^2, compute the secant slope from x = -1 to x = 1.

[f(p+h)-f(p)]/h = 2[(p+h)^2-p^2]/h = 2(2p+h) = 0.

L38-P023 | For f(x) = 3x^2, compute the secant slope from x = 0 to x = 3.

[f(p+h)-f(p)]/h = 3[(p+h)^2-p^2]/h = 3(2p+h) = 9.

L38-P024 | For f(x) = 4x^2, compute the secant slope from x = 1 to x = 2.

[f(p+h)-f(p)]/h = 4[(p+h)^2-p^2]/h = 4(2p+h) = 12.

L38-P025 | For f(x) = 1x^2, compute the secant slope from x = 2 to x = 4.

[f(p+h)-f(p)]/h = 1[(p+h)^2-p^2]/h = 1(2p+h) = 6.

L38-P026 | For f(x) = 2x^2, compute the secant slope from x = -2 to x = 1.

[f(p+h)-f(p)]/h = 2[(p+h)^2-p^2]/h = 2(2p+h) = -2.

L38-P027 | For f(x) = 3x^2, compute the secant slope from x = -1 to x = 0.

[f(p+h)-f(p)]/h = 3[(p+h)^2-p^2]/h = 3(2p+h) = -3.

L38-P028 | For f(x) = 4x^2, compute the secant slope from x = 0 to x = 2.

[f(p+h)-f(p)]/h = 4[(p+h)^2-p^2]/h = 4(2p+h) = 8.

L38-P029 | For f(x) = 1x^2, compute the secant slope from x = 1 to x = 4.

[f(p+h)-f(p)]/h = 1[(p+h)^2-p^2]/h = 1(2p+h) = 5.

L38-P030 | For f(x) = 2x^2, compute the secant slope from x = 2 to x = 3.

[f(p+h)-f(p)]/h = 2[(p+h)^2-p^2]/h = 2(2p+h) = 10.

L38-P031 | For f(x) = 3x^2, compute the secant slope from x = -2 to x = 0.

[f(p+h)-f(p)]/h = 3[(p+h)^2-p^2]/h = 3(2p+h) = -6.

L38-P032 | For f(x) = 4x^2, compute the secant slope from x = -1 to x = 2.

[f(p+h)-f(p)]/h = 4[(p+h)^2-p^2]/h = 4(2p+h) = 4.

L38-P033 | For f(x) = 1x^2, compute the secant slope from x = 0 to x = 1.

[f(p+h)-f(p)]/h = 1[(p+h)^2-p^2]/h = 1(2p+h) = 1.

L38-P034 | For f(x) = 2x^2, compute the secant slope from x = 1 to x = 3.

[f(p+h)-f(p)]/h = 2[(p+h)^2-p^2]/h = 2(2p+h) = 8.

L38-P035 | For f(x) = 3x^2, compute the secant slope from x = 2 to x = 5.

[f(p+h)-f(p)]/h = 3[(p+h)^2-p^2]/h = 3(2p+h) = 21.

L38-P036 | For f(x) = 4x^2, compute the secant slope from x = -2 to x = -1.

[f(p+h)-f(p)]/h = 4[(p+h)^2-p^2]/h = 4(2p+h) = -12.

L38-P037 | For f(x) = 1x^2, compute the secant slope from x = -1 to x = 1.

[f(p+h)-f(p)]/h = 1[(p+h)^2-p^2]/h = 1(2p+h) = 0.

L38-P038 | For f(x) = 2x^2, compute the secant slope from x = 0 to x = 3.

[f(p+h)-f(p)]/h = 2[(p+h)^2-p^2]/h = 2(2p+h) = 6.

L38-P039 | For f(x) = 3x^2, compute the secant slope from x = 1 to x = 2.

[f(p+h)-f(p)]/h = 3[(p+h)^2-p^2]/h = 3(2p+h) = 9.

L38-P040 | For f(x) = 4x^2, compute the secant slope from x = 2 to x = 4.

[f(p+h)-f(p)]/h = 4[(p+h)^2-p^2]/h = 4(2p+h) = 24.

3/5 | Challenger

L38-P041 | Using the difference quotient limit, find f’(-2) for f(x) = 1x^2 - 3x.

[f(p+h)-f(p)]/h = 1(2p+h) - 3; as h→0: 1×2×(-2) - 3 = -7.

L38-P042 | Using the difference quotient limit, find f’(-1) for f(x) = 2x^2 - 2x.

[f(p+h)-f(p)]/h = 2(2p+h) - 2; as h→0: 2×2×(-1) - 2 = -6.

L38-P043 | Using the difference quotient limit, find f’(0) for f(x) = 3x^2 - 1x.

[f(p+h)-f(p)]/h = 3(2p+h) - 1; as h→0: 3×2×(0) - 1 = -1.

L38-P044 | Using the difference quotient limit, find f’(1) for f(x) = 4x^2 + 0x.

[f(p+h)-f(p)]/h = 4(2p+h) + 0; as h→0: 4×2×(1) + 0 = 8.

L38-P045 | Using the difference quotient limit, find f’(2) for f(x) = 1x^2 + 1x.

[f(p+h)-f(p)]/h = 1(2p+h) + 1; as h→0: 1×2×(2) + 1 = 5.

L38-P046 | Using the difference quotient limit, find f’(-2) for f(x) = 2x^2 + 2x.

[f(p+h)-f(p)]/h = 2(2p+h) + 2; as h→0: 2×2×(-2) + 2 = -6.

L38-P047 | Using the difference quotient limit, find f’(-1) for f(x) = 3x^2 + 3x.

[f(p+h)-f(p)]/h = 3(2p+h) + 3; as h→0: 3×2×(-1) + 3 = -3.

L38-P048 | Using the difference quotient limit, find f’(0) for f(x) = 4x^2 - 3x.

[f(p+h)-f(p)]/h = 4(2p+h) - 3; as h→0: 4×2×(0) - 3 = -3.

L38-P049 | Using the difference quotient limit, find f’(1) for f(x) = 1x^2 - 2x.

[f(p+h)-f(p)]/h = 1(2p+h) - 2; as h→0: 1×2×(1) - 2 = 0.

L38-P050 | Using the difference quotient limit, find f’(2) for f(x) = 2x^2 - 1x.

[f(p+h)-f(p)]/h = 2(2p+h) - 1; as h→0: 2×2×(2) - 1 = 7.

L38-P051 | Using the difference quotient limit, find f’(-2) for f(x) = 3x^2 + 0x.

[f(p+h)-f(p)]/h = 3(2p+h) + 0; as h→0: 3×2×(-2) + 0 = -12.

L38-P052 | Using the difference quotient limit, find f’(-1) for f(x) = 4x^2 + 1x.

[f(p+h)-f(p)]/h = 4(2p+h) + 1; as h→0: 4×2×(-1) + 1 = -7.

L38-P053 | Using the difference quotient limit, find f’(0) for f(x) = 1x^2 + 2x.

[f(p+h)-f(p)]/h = 1(2p+h) + 2; as h→0: 1×2×(0) + 2 = 2.

L38-P054 | Using the difference quotient limit, find f’(1) for f(x) = 2x^2 + 3x.

[f(p+h)-f(p)]/h = 2(2p+h) + 3; as h→0: 2×2×(1) + 3 = 7.

L38-P055 | Using the difference quotient limit, find f’(2) for f(x) = 3x^2 - 3x.

[f(p+h)-f(p)]/h = 3(2p+h) - 3; as h→0: 3×2×(2) - 3 = 9.

L38-P056 | Using the difference quotient limit, find f’(-2) for f(x) = 4x^2 - 2x.

[f(p+h)-f(p)]/h = 4(2p+h) - 2; as h→0: 4×2×(-2) - 2 = -18.

L38-P057 | Using the difference quotient limit, find f’(-1) for f(x) = 1x^2 - 1x.

[f(p+h)-f(p)]/h = 1(2p+h) - 1; as h→0: 1×2×(-1) - 1 = -3.

L38-P058 | Using the difference quotient limit, find f’(0) for f(x) = 2x^2 + 0x.

[f(p+h)-f(p)]/h = 2(2p+h) + 0; as h→0: 2×2×(0) + 0 = 0.

L38-P059 | Using the difference quotient limit, find f’(1) for f(x) = 3x^2 + 1x.

[f(p+h)-f(p)]/h = 3(2p+h) + 1; as h→0: 3×2×(1) + 1 = 7.

L38-P060 | Using the difference quotient limit, find f’(2) for f(x) = 4x^2 + 2x.

[f(p+h)-f(p)]/h = 4(2p+h) + 2; as h→0: 4×2×(2) + 2 = 18.

4/5 | Expert

L38-P061 | f(x) = 1x^2 - 3x. Find the tangent-line equation at x = -2.

f(-2) = 10; f’(-2) = -7. Tangent: y = 10 + (-7)(x - (-2)).

L38-P062 | f(x) = 2x^2 - 2x. Find the tangent-line equation at x = -1.

f(-1) = 4; f’(-1) = -6. Tangent: y = 4 + (-6)(x - (-1)).

L38-P063 | f(x) = 3x^2 - 1x. Find the tangent-line equation at x = 0.

f(0) = 0; f’(0) = -1. Tangent: y = 0 + (-1)(x - (0)).

L38-P064 | f(x) = 4x^2 + 0x. Find the tangent-line equation at x = 1.

f(1) = 4; f’(1) = 8. Tangent: y = 4 + (8)(x - (1)).

L38-P065 | f(x) = 1x^2 + 1x. Find the tangent-line equation at x = 2.

f(2) = 6; f’(2) = 5. Tangent: y = 6 + (5)(x - (2)).

L38-P066 | f(x) = 2x^2 + 2x. Find the tangent-line equation at x = -2.

f(-2) = 4; f’(-2) = -6. Tangent: y = 4 + (-6)(x - (-2)).

L38-P067 | f(x) = 3x^2 + 3x. Find the tangent-line equation at x = -1.

f(-1) = 0; f’(-1) = -3. Tangent: y = 0 + (-3)(x - (-1)).

L38-P068 | f(x) = 4x^2 - 3x. Find the tangent-line equation at x = 0.

f(0) = 0; f’(0) = -3. Tangent: y = 0 + (-3)(x - (0)).

L38-P069 | f(x) = 1x^2 - 2x. Find the tangent-line equation at x = 1.

f(1) = -1; f’(1) = 0. Tangent: y = -1 + (0)(x - (1)).

L38-P070 | f(x) = 2x^2 - 1x. Find the tangent-line equation at x = 2.

f(2) = 6; f’(2) = 7. Tangent: y = 6 + (7)(x - (2)).

L38-P071 | f(x) = 3x^2 + 0x. Find the tangent-line equation at x = -2.

f(-2) = 12; f’(-2) = -12. Tangent: y = 12 + (-12)(x - (-2)).

L38-P072 | f(x) = 4x^2 + 1x. Find the tangent-line equation at x = -1.

f(-1) = 3; f’(-1) = -7. Tangent: y = 3 + (-7)(x - (-1)).

L38-P073 | f(x) = 1x^2 + 2x. Find the tangent-line equation at x = 0.

f(0) = 0; f′(0) = 2. Tangent: y = 0 + (2)(x - (0)).

L38-P074 | f(x) = 2x^2 + 3x. Find the tangent-line equation at x = 1.

f(1) = 5; f′(1) = 7. Tangent: y = 5 + (7)(x - (1)).

L38-P075 | f(x) = 3x^2 - 3x. Find the tangent-line equation at x = 2.

f(2) = 6; f′(2) = 9. Tangent: y = 6 + (9)(x - (2)).

L38-P076 | f(x) = 4x^2 - 2x. Find the tangent-line equation at x = -2.

f(-2) = 20; f′(-2) = -18. Tangent: y = 20 + (-18)(x - (-2)).

L38-P077 | f(x) = 1x^2 - 1x. Find the tangent-line equation at x = -1.

f(-1) = 2; f′(-1) = -3. Tangent: y = 2 + (-3)(x - (-1)).

L38-P078 | f(x) = 2x^2 + 0x. Find the tangent-line equation at x = 0.

f(0) = 0; f′(0) = 0. Tangent: y = 0 + (0)(x - (0)).

L38-P079 | f(x) = 3x^2 + 1x. Find the tangent-line equation at x = 1.

f(1) = 4; f′(1) = 7. Tangent: y = 4 + (7)(x - (1)).

L38-P080 | f(x) = 4x^2 + 2x. Find the tangent-line equation at x = 2.

f(2) = 20; f′(2) = 18. Tangent: y = 20 + (18)(x - (2)).

5/5 | Ninja

L38-P081 | From the definition, differentiate f(x) = 1x^3 - 3x^2 - 1x at x = -2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×1×(-2)^2 + 2×(-3)×(-2) + (-1) = 23. y = -18 + (23)(x - (-2)).

L38-P082 | From the definition, differentiate f(x) = 2x^3 - 2x^2 + 0x at x = -1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×2×(-1)^2 + 2×(-2)×(-1) + (0) = 10. y = -4 + (10)(x - (-1)).

L38-P083 | From the definition, differentiate f(x) = 3x^3 - 1x^2 + 1x at x = 0; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×3×(0)^2 + 2×(-1)×(0) + (1) = 1. y = 0 + (1)(x - (0)).

L38-P084 | From the definition, differentiate f(x) = 4x^3 + 0x^2 + 2x at x = 1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×4×(1)^2 + 2×(0)×(1) + (2) = 14. y = 6 + (14)(x - (1)).

L38-P085 | From the definition, differentiate f(x) = 1x^3 + 1x^2 - 1x at x = 2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×1×(2)^2 + 2×(1)×(2) + (-1) = 15. y = 10 + (15)(x - (2)).

L38-P086 | From the definition, differentiate f(x) = 2x^3 + 2x^2 + 0x at x = -2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×2×(-2)^2 + 2×(2)×(-2) + (0) = 16. y = -8 + (16)(x - (-2)).

L38-P087 | From the definition, differentiate f(x) = 3x^3 + 3x^2 + 1x at x = -1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×3×(-1)^2 + 2×(3)×(-1) + (1) = 4. y = -1 + (4)(x - (-1)).

L38-P088 | From the definition, differentiate f(x) = 4x^3 - 3x^2 + 2x at x = 0; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×4×(0)^2 + 2×(-3)×(0) + (2) = 2. y = 0 + (2)(x - (0)).

L38-P089 | From the definition, differentiate f(x) = 1x^3 - 2x^2 - 1x at x = 1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×1×(1)^2 + 2×(-2)×(1) + (-1) = -2. y = -2 + (-2)(x - (1)).

L38-P090 | From the definition, differentiate f(x) = 2x^3 - 1x^2 + 0x at x = 2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×2×(2)^2 + 2×(-1)×(2) + (0) = 20. y = 12 + (20)(x - (2)).

L38-P091 | From the definition, differentiate f(x) = 3x^3 + 0x^2 + 1x at x = -2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×3×(-2)^2 + 2×(0)×(-2) + (1) = 37. y = -26 + (37)(x - (-2)).

L38-P092 | From the definition, differentiate f(x) = 4x^3 + 1x^2 + 2x at x = -1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×4×(-1)^2 + 2×(1)×(-1) + (2) = 12. y = -5 + (12)(x - (-1)).

L38-P093 | From the definition, differentiate f(x) = 1x^3 + 2x^2 - 1x at x = 0; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×1×(0)^2 + 2×(2)×(0) + (-1) = -1. y = 0 + (-1)(x - (0)).

L38-P094 | From the definition, differentiate f(x) = 2x^3 + 3x^2 + 0x at x = 1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×2×(1)^2 + 2×(3)×(1) + (0) = 12. y = 5 + (12)(x - (1)).

L38-P095 | From the definition, differentiate f(x) = 3x^3 - 3x^2 + 1x at x = 2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×3×(2)^2 + 2×(-3)×(2) + (1) = 25. y = 14 + (25)(x - (2)).

L38-P096 | From the definition, differentiate f(x) = 4x^3 - 2x^2 + 2x at x = -2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×4×(-2)^2 + 2×(-2)×(-2) + (2) = 58. y = -44 + (58)(x - (-2)).

L38-P097 | From the definition, differentiate f(x) = 1x^3 - 1x^2 - 1x at x = -1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×1×(-1)^2 + 2×(-1)×(-1) + (-1) = 4. y = -1 + (4)(x - (-1)).

L38-P098 | From the definition, differentiate f(x) = 2x^3 + 0x^2 + 0x at x = 0; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×2×(0)^2 + 2×(0)×(0) + (0) = 0. y = 0 + (0)(x - (0)).

L38-P099 | From the definition, differentiate f(x) = 3x^3 + 1x^2 + 1x at x = 1; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×3×(1)^2 + 2×(1)×(1) + (1) = 12. y = 5 + (12)(x - (1)).

L38-P100 | From the definition, differentiate f(x) = 4x^3 + 2x^2 + 2x at x = 2; find its tangent.

Expand (p+h)^3 and (p+h)^2, subtract f(p), divide by h; the limit is 3×4×(2)^2 + 2×(2)×(2) + (2) = 58. y = 44 + (58)(x - (2)).

L39 | Differentiation rules

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1/5 | Apprentice

L39-P001 | Differentiate f(x) = 1x^2.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 1×2 x^1 = 2x^1.

L39-P002 | Differentiate f(x) = 2x^3.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 2×3 x^2 = 6x^2.

L39-P003 | Differentiate f(x) = 3x^4.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 3×4 x^3 = 12x^3.

L39-P004 | Differentiate f(x) = 4x^5.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 4×5 x^4 = 20x^4.

L39-P005 | Differentiate f(x) = 5x^6.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 5×6 x^5 = 30x^5.

L39-P006 | Differentiate f(x) = 1x^2.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 1×2 x^1 = 2x^1.

L39-P007 | Differentiate f(x) = 2x^3.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 2×3 x^2 = 6x^2.

L39-P008 | Differentiate f(x) = 3x^4.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 3×4 x^3 = 12x^3.

L39-P009 | Differentiate f(x) = 4x^5.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 4×5 x^4 = 20x^4.

L39-P010 | Differentiate f(x) = 5x^6.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 5×6 x^5 = 30x^5.

L39-P011 | Differentiate f(x) = 1x^2.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 1×2 x^1 = 2x^1.

L39-P012 | Differentiate f(x) = 2x^3.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 2×3 x^2 = 6x^2.

L39-P013 | Differentiate f(x) = 3x^4.

Power rule: D(x^n)=n x^(n-1), hence f'(x) = 3×4 x^3 = 12x^3.

L39-P014 | Differentiate f(x) = 4x^5.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 4×5 x^4 = 20x^4.

L39-P015 | Differentiate f(x) = 5x^6.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 5×6 x^5 = 30x^5.

L39-P016 | Differentiate f(x) = 1x^2.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 1×2 x^1 = 2x^1.

L39-P017 | Differentiate f(x) = 2x^3.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 2×3 x^2 = 6x^2.

L39-P018 | Differentiate f(x) = 3x^4.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 3×4 x^3 = 12x^3.

L39-P019 | Differentiate f(x) = 4x^5.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 4×5 x^4 = 20x^4.

L39-P020 | Differentiate f(x) = 5x^6.

Power rule: D(x^n)=n x^(n-1), hence f’(x) = 5×6 x^5 = 30x^5.

2/5 | Practitioner

L39-P021 | f(x) = 1x^2 - 3x + 1. Find f’(-2).

f’(x) = 2x - 3 (constant derivative is 0). f’(-2) = -7.

L39-P022 | f(x) = 2x^2 - 2x + 2. Find f’(-1).

f’(x) = 4x - 2 (constant derivative is 0). f’(-1) = -6.

L39-P023 | f(x) = 3x^2 - 1x + 3. Find f’(0).

f’(x) = 6x - 1 (constant derivative is 0). f’(0) = -1.

L39-P024 | f(x) = 4x^2 + 0x + 4. Find f’(1).

f’(x) = 8x + 0 (constant derivative is 0). f’(1) = 8.

L39-P025 | f(x) = 5x^2 + 1x + 1. Find f’(2).

f’(x) = 10x + 1 (constant derivative is 0). f’(2) = 21.

L39-P026 | f(x) = 1x^2 + 2x + 2. Find f’(-2).

f’(x) = 2x + 2 (constant derivative is 0). f’(-2) = -2.

L39-P027 | f(x) = 2x^2 + 3x + 3. Find f’(-1).

f’(x) = 4x + 3 (constant derivative is 0). f’(-1) = -1.

L39-P028 | f(x) = 3x^2 - 3x + 4. Find f’(0).

f’(x) = 6x - 3 (constant derivative is 0). f’(0) = -3.

L39-P029 | f(x) = 4x^2 - 2x + 1. Find f’(1).

f’(x) = 8x - 2 (constant derivative is 0). f’(1) = 6.

L39-P030 | f(x) = 5x^2 - 1x + 2. Find f’(2).

f’(x) = 10x - 1 (constant derivative is 0). f’(2) = 19.

L39-P031 | f(x) = 1x^2 + 0x + 3. Find f’(-2).

f’(x) = 2x + 0 (constant derivative is 0). f’(-2) = -4.

L39-P032 | f(x) = 2x^2 + 1x + 4. Find f’(-1).

f’(x) = 4x + 1 (constant derivative is 0). f’(-1) = -3.

L39-P033 | f(x) = 3x^2 + 2x + 1. Find f’(0).

f’(x) = 6x + 2 (constant derivative is 0). f’(0) = 2.

L39-P034 | f(x) = 4x^2 + 3x + 2. Find f’(1).

f’(x) = 8x + 3 (constant derivative is 0). f’(1) = 11.

L39-P035 | f(x) = 5x^2 - 3x + 3. Find f’(2).

f’(x) = 10x - 3 (constant derivative is 0). f’(2) = 17.

L39-P036 | f(x) = 1x^2 - 2x + 4. Find f’(-2).

f’(x) = 2x - 2 (constant derivative is 0). f’(-2) = -6.

L39-P037 | f(x) = 2x^2 - 1x + 1. Find f’(-1).

f’(x) = 4x - 1 (constant derivative is 0). f’(-1) = -5.

L39-P038 | f(x) = 3x^2 + 0x + 2. Find f’(0).

f’(x) = 6x + 0 (constant derivative is 0). f’(0) = 0.

L39-P039 | f(x) = 4x^2 + 1x + 3. Find f’(1).

f’(x) = 8x + 1 (constant derivative is 0). f’(1) = 9.

L39-P040 | f(x) = 5x^2 + 2x + 4. Find f’(2).

f’(x) = 10x + 2 (constant derivative is 0). f’(2) = 22.

3/5 | Challenger

L39-P041 | Differentiate f(x) = (1x - 3)(1x - 3).

Set u = 1x - 3, v = 1x - 3. f’ = 1(1x - 3) + (1x - 3)1 = 2*x - 6.

L39-P042 | Differentiate f(x) = (2x - 2)(2x + 0).

Set u = 2x - 2, v = 2x + 0. f’ = 2(2x + 0) + (2x - 2)2 = 8*x - 4.

L39-P043 | Differentiate f(x) = (3x - 1)(3x + 3).

Set u = 3x - 1, v = 3x + 3. f’ = 3(3x + 3) + (3x - 1)3 = 18*x + 6.

L39-P044 | Differentiate f(x) = (4x + 0)(4x - 1).

Set u = 4x + 0, v = 4x - 1. f’ = 4(4x - 1) + (4x + 0)4 = 32*x - 4.

L39-P045 | Differentiate f(x) = (5x + 1)(1x + 2).

Set u = 5x + 1, v = 1x + 2. f’ = 5(1x + 2) + (5x + 1)1 = 10*x + 11.

L39-P046 | Differentiate f(x) = (1x + 2)(2x - 2).

Set u = 1x + 2, v = 2x - 2. f’ = 1(2x - 2) + (1x + 2)2 = 4*x + 2.

L39-P047 | Differentiate f(x) = (2x + 3)(3x + 1).

Set u = 2x + 3, v = 3x + 1. f’ = 2(3x + 1) + (2x + 3)3 = 12*x + 11.

L39-P048 | Differentiate f(x) = (3x - 3)(4x - 3).

Set u = 3x - 3, v = 4x - 3. f’ = 3(4x - 3) + (3x - 3)4 = 24*x - 21.

L39-P049 | Differentiate f(x) = (4x - 2)(1x + 0).

Set u = 4x - 2, v = 1x + 0. f’ = 4(1x + 0) + (4x - 2)1 = 8*x - 2.

L39-P050 | Differentiate f(x) = (5x - 1)(2x + 3).

Set u = 5x - 1, v = 2x + 3. f’ = 5(2x + 3) + (5x - 1)2 = 20*x + 13.

L39-P051 | Differentiate f(x) = (1x + 0)(3x - 1).

Set u = 1x + 0, v = 3x - 1. f’ = 1(3x - 1) + (1x + 0)3 = 6*x - 1.

L39-P052 | Differentiate f(x) = (2x + 1)(4x + 2).

Set u = 2x + 1, v = 4x + 2. f’ = 2(4x + 2) + (2x + 1)4 = 16*x + 8.

L39-P053 | Differentiate f(x) = (3x + 2)(1x - 2).

Set u = 3x + 2, v = 1x - 2. f’ = 3(1x - 2) + (3x + 2)1 = 6*x - 4.

L39-P054 | Differentiate f(x) = (4x + 3)(2x + 1).

Set u = 4x + 3, v = 2x + 1. f’ = 4(2x + 1) + (4x + 3)2 = 16*x + 10.

L39-P055 | Differentiate f(x) = (5x - 3)(3x - 3).

Set u = 5x - 3, v = 3x - 3. f’ = 5(3x - 3) + (5x - 3)3 = 30*x - 24.

L39-P056 | Differentiate f(x) = (1x - 2)(4x + 0).

Set u = 1x - 2, v = 4x + 0. f’ = 1(4x + 0) + (1x - 2)4 = 8*x - 8.

L39-P057 | Differentiate f(x) = (2x - 1)(1x + 3).

Set u = 2x - 1, v = 1x + 3. f’ = 2(1x + 3) + (2x - 1)1 = 4*x + 5.

L39-P058 | Differentiate f(x) = (3x + 0)(2x - 1).

Set u = 3x + 0, v = 2x - 1. f’ = 3(2x - 1) + (3x + 0)2 = 12*x - 3.

L39-P059 | Differentiate f(x) = (4x + 1)(3x + 2).

Set u = 4x + 1, v = 3x + 2. f’ = 4(3x + 2) + (4x + 1)3 = 24*x + 11.

L39-P060 | Differentiate f(x) = (5x + 2)(4x - 2).

Set u = 5x + 2, v = 4x - 2. f’ = 5(4x - 2) + (5x + 2)4 = 40*x - 2.

4/5 | Expert

L39-P061 | Differentiate f(x) = (x^2 + 2)/(x - 1), stating its domain.

Domain x ≠ 1. Quotient rule: f’ = [2x(x - 1) - (x^2 + 2)]/(x - 1)^2 = (x^2 - 2x - 2)/(x - 1)^2.

L39-P062 | Differentiate f(x) = (x^2 + 3)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 3)]/(x - 2)^2 = (x^2 - 4x - 3)/(x - 2)^2.

L39-P063 | Differentiate f(x) = (x^2 + 4)/(x - 3), stating its domain.

Domain x ≠ 3. Quotient rule: f’ = [2x(x - 3) - (x^2 + 4)]/(x - 3)^2 = (x^2 - 6x - 4)/(x - 3)^2.

L39-P064 | Differentiate f(x) = (x^2 + 5)/(x - 4), stating its domain.

Domain x ≠ 4. Quotient rule: f’ = [2x(x - 4) - (x^2 + 5)]/(x - 4)^2 = (x^2 - 8x - 5)/(x - 4)^2.

L39-P065 | Differentiate f(x) = (x^2 + 6)/(x - 5), stating its domain.

Domain x ≠ 5. Quotient rule: f’ = [2x(x - 5) - (x^2 + 6)]/(x - 5)^2 = (x^2 - 10x - 6)/(x - 5)^2.

L39-P066 | Differentiate f(x) = (x^2 + 7)/(x - 1), stating its domain.

Domain x ≠ 1. Quotient rule: f’ = [2x(x - 1) - (x^2 + 7)]/(x - 1)^2 = (x^2 - 2x - 7)/(x - 1)^2.

L39-P067 | Differentiate f(x) = (x^2 + 2)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 2)]/(x - 2)^2 = (x^2 - 4x - 2)/(x - 2)^2.

L39-P068 | Differentiate f(x) = (x^2 + 3)/(x - 3), stating its domain.

Domain x ≠ 3. Quotient rule: f’ = [2x(x - 3) - (x^2 + 3)]/(x - 3)^2 = (x^2 - 6x - 3)/(x - 3)^2.

L39-P069 | Differentiate f(x) = (x^2 + 4)/(x - 4), stating its domain.

Domain x ≠ 4. Quotient rule: f’ = [2x(x - 4) - (x^2 + 4)]/(x - 4)^2 = (x^2 - 8x - 4)/(x - 4)^2.

L39-P070 | Differentiate f(x) = (x^2 + 5)/(x - 5), stating its domain.

Domain x ≠ 5. Quotient rule: f’ = [2x(x - 5) - (x^2 + 5)]/(x - 5)^2 = (x^2 - 10x - 5)/(x - 5)^2.

L39-P071 | Differentiate f(x) = (x^2 + 6)/(x - 1), stating its domain.

Domain x ≠ 1. Quotient rule: f’ = [2x(x - 1) - (x^2 + 6)]/(x - 1)^2 = (x^2 - 2x - 6)/(x - 1)^2.

L39-P072 | Differentiate f(x) = (x^2 + 7)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 7)]/(x - 2)^2 = (x^2 - 4x - 7)/(x - 2)^2.

L39-P073 | Differentiate f(x) = (x^2 + 2)/(x - 3), stating its domain.

Domain x ≠ 3. Quotient rule: f’ = [2x(x - 3) - (x^2 + 2)]/(x - 3)^2 = (x^2 - 6x - 2)/(x - 3)^2.

L39-P074 | Differentiate f(x) = (x^2 + 3)/(x - 4), stating its domain.

Domain x ≠ 4. Quotient rule: f’ = [2x(x - 4) - (x^2 + 3)]/(x - 4)^2 = (x^2 - 8x - 3)/(x - 4)^2.

L39-P075 | Differentiate f(x) = (x^2 + 4)/(x - 5), stating its domain.

Domain x ≠ 5. Quotient rule: f’ = [2x(x - 5) - (x^2 + 4)]/(x - 5)^2 = (x^2 - 10x - 4)/(x - 5)^2.

L39-P076 | Differentiate f(x) = (x^2 + 5)/(x - 1), stating its domain.

Domain x ≠ 1. Quotient rule: f’ = [2x(x - 1) - (x^2 + 5)]/(x - 1)^2 = (x^2 - 2x - 5)/(x - 1)^2.

L39-P077 | Differentiate f(x) = (x^2 + 6)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 6)]/(x - 2)^2 = (x^2 - 4x - 6)/(x - 2)^2.

L39-P078 | Differentiate f(x) = (x^2 + 7)/(x - 3), stating its domain.

Domain x ≠ 3. Quotient rule: f’ = [2x(x - 3) - (x^2 + 7)]/(x - 3)^2 = (x^2 - 6x - 7)/(x - 3)^2.

L39-P079 | Differentiate f(x) = (x^2 + 2)/(x - 4), stating its domain.

Domain x ≠ 4. Quotient rule: f’ = [2x(x - 4) - (x^2 + 2)]/(x - 4)^2 = (x^2 - 8x - 2)/(x - 4)^2.

L39-P080 | Differentiate f(x) = (x^2 + 3)/(x - 5), stating its domain.

Domain x ≠ 5. Quotient rule: f’ = [2x(x - 5) - (x^2 + 3)]/(x - 5)^2 = (x^2 - 10x - 3)/(x - 5)^2.

5/5 | Ninja

L39-P081 | Use the chain rule to differentiate f(x) = (1x + 0)^2, then find f’(1).

If u(x) = 1x + 0, f’ = 2u^1u’ = 2(1x + 0)^1. Thus f’(1) = 2.

L39-P082 | Use the chain rule to differentiate f(x) = (2x + 1)^3, then find f’(2).

If u(x) = 2x + 1, f’ = 3u^2u’ = 6(2x + 1)^2. Thus f’(2) = 150.

L39-P083 | Use the chain rule to differentiate f(x) = (3x + 2)^4, then find f’(3).

If u(x) = 3x + 2, f’ = 4u^3u’ = 12(3x + 2)^3. Thus f’(3) = 15972.

L39-P084 | Use the chain rule to differentiate f(x) = (4x + 3)^5, then find f’(1).

If u(x) = 4x + 3, f’ = 5u^4u’ = 20(4x + 3)^4. Thus f’(1) = 48020.

L39-P085 | Use the chain rule to differentiate f(x) = (5x + 0)^2, then find f’(2).

If u(x) = 5x + 0, f’ = 2u^1u’ = 10(5x + 0)^1. Thus f’(2) = 100.

L39-P086 | Use the chain rule to differentiate f(x) = (1x + 1)^3, then find f’(3).

If u(x) = 1x + 1, f’ = 3u^2u’ = 3(1x + 1)^2. Thus f’(3) = 48.

L39-P087 | Use the chain rule to differentiate f(x) = (2x + 2)^4, then find f’(1).

If u(x) = 2x + 2, f’ = 4u^3u’ = 8(2x + 2)^3. Thus f’(1) = 512.

L39-P088 | Use the chain rule to differentiate f(x) = (3x + 3)^5, then find f’(2).

If u(x) = 3x + 3, f’ = 5u^4u’ = 15(3x + 3)^4. Thus f’(2) = 98415.

L39-P089 | Use the chain rule to differentiate f(x) = (4x + 0)^2, then find f’(3).

If u(x) = 4x + 0, f’ = 2u^1u’ = 8(4x + 0)^1. Thus f’(3) = 96.

L39-P090 | Use the chain rule to differentiate f(x) = (5x + 1)^3, then find f’(1).

If u(x) = 5x + 1, f’ = 3u^2u’ = 15(5x + 1)^2. Thus f’(1) = 540.

L39-P091 | Use the chain rule to differentiate f(x) = (1x + 2)^4, then find f’(2).

If u(x) = 1x + 2, f’ = 4u^3u’ = 4(1x + 2)^3. Thus f’(2) = 256.

L39-P092 | Use the chain rule to differentiate f(x) = (2x + 3)^5, then find f’(3).

If u(x) = 2x + 3, f’ = 5u^4u’ = 10(2x + 3)^4. Thus f’(3) = 65610.

L39-P093 | Use the chain rule to differentiate f(x) = (3x + 0)^2, then find f’(1).

If u(x) = 3x + 0, f’ = 2u^1u’ = 6(3x + 0)^1. Thus f’(1) = 18.

L39-P094 | Use the chain rule to differentiate f(x) = (4x + 1)^3, then find f’(2).

If u(x) = 4x + 1, f’ = 3u^2u’ = 12(4x + 1)^2. Thus f’(2) = 972.

L39-P095 | Use the chain rule to differentiate f(x) = (5x + 2)^4, then find f’(3).

If u(x) = 5x + 2, f’ = 4u^3u’ = 20(5x + 2)^3. Thus f’(3) = 98260.

L39-P096 | Use the chain rule to differentiate f(x) = (1x + 3)^5, then find f’(1).

If u(x) = 1x + 3, f’ = 5u^4u’ = 5(1x + 3)^4. Thus f’(1) = 1280.

L39-P097 | Use the chain rule to differentiate f(x) = (2x + 0)^2, then find f’(2).

If u(x) = 2x + 0, f’ = 2u^1u’ = 4(2x + 0)^1. Thus f’(2) = 16.

L39-P098 | Use the chain rule to differentiate f(x) = (3x + 1)^3, then find f’(3).

If u(x) = 3x + 1, f’ = 3u^2u’ = 9(3x + 1)^2. Thus f’(3) = 900.

L39-P099 | Use the chain rule to differentiate f(x) = (4x + 2)^4, then find f’(1).

If u(x) = 4x + 2, f’ = 4u^3u’ = 16(4x + 2)^3. Thus f’(1) = 3456.

L39-P100 | Use the chain rule to differentiate f(x) = (5x + 3)^5, then find f’(2).

If u(x) = 5x + 3, f’ = 5u^4u’ = 25(5x + 3)^4. Thus f’(2) = 714025.

L40 | The natural exponential function

Source: PDF pages 14, 15

1/5 | Apprentice

L40-P001 | Simplify e^2 × e^1 / e^0.

Add then subtract exponents: e^(2+1-0) = e^3.

L40-P002 | Simplify e^3 × e^2 / e^1.

Add then subtract exponents: e^(3+2-1) = e^4.

L40-P003 | Simplify e^4 × e^3 / e^2.

Add then subtract exponents: e^(4+3-2) = e^5.

L40-P004 | Simplify e^5 × e^4 / e^3.

Add then subtract exponents: e^(5+4-3) = e^6.

L40-P005 | Simplify e^6 × e^1 / e^4.

Add then subtract exponents: e^(6+1-4) = e^3.

L40-P006 | Simplify e^2 × e^2 / e^0.

Add then subtract exponents: e^(2+2-0) = e^4.

L40-P007 | Simplify e^3 × e^3 / e^1.

Add then subtract exponents: e^(3+3-1) = e^5.

L40-P008 | Simplify e^4 × e^4 / e^2.

Add then subtract exponents: e^(4+4-2) = e^6.

L40-P009 | Simplify e^5 × e^1 / e^3.

Add then subtract exponents: e^(5+1-3) = e^3.

L40-P010 | Simplify e^6 × e^2 / e^4.

Add then subtract exponents: e^(6+2-4) = e^4.

L40-P011 | Simplify e^2 × e^3 / e^0.

Add then subtract exponents: e^(2+3-0) = e^5.

L40-P012 | Simplify e^3 × e^4 / e^1.

Add then subtract exponents: e^(3+4-1) = e^6.

L40-P013 | Simplify e^4 × e^1 / e^2.

Add then subtract exponents: e^(4+1-2) = e^3.

L40-P014 | Simplify e^5 × e^2 / e^3.

Add then subtract exponents: e^(5+2-3) = e^4.

L40-P015 | Simplify e^6 × e^3 / e^4.

Add then subtract exponents: e^(6+3-4) = e^5.

L40-P016 | Simplify e^2 × e^4 / e^0.

Add then subtract exponents: e^(2+4-0) = e^6.

L40-P017 | Simplify e^3 × e^1 / e^1.

Add then subtract exponents: e^(3+1-1) = e^3.

L40-P018 | Simplify e^4 × e^2 / e^2.

Add then subtract exponents: e^(4+2-2) = e^4.

L40-P019 | Simplify e^5 × e^3 / e^3.

Add then subtract exponents: e^(5+3-3) = e^5.

L40-P020 | Simplify e^6 × e^4 / e^4.

Add then subtract exponents: e^(6+4-4) = e^6.

2/5 | Practitioner

L40-P021 | Compute f(0) and f’(0) for f(x) = 2e^x + 1.

e^0=1, so f(0) = 2+1 = 3. Its derivative is 2e^x, so f’(0) = 2.

L40-P022 | Compute f(0) and f’(0) for f(x) = 3e^x + 2.

e^0=1, so f(0) = 3+2 = 5. Its derivative is 3e^x, so f’(0) = 3.

L40-P023 | Compute f(0) and f’(0) for f(x) = 4e^x + 3.

e^0=1, so f(0) = 4+3 = 7. Its derivative is 4e^x, so f’(0) = 4.

L40-P024 | Compute f(0) and f’(0) for f(x) = 5e^x + 4.

e^0=1, so f(0) = 5+4 = 9. Its derivative is 5e^x, so f’(0) = 5.

L40-P025 | Compute f(0) and f′(0) for f(x) = 6e^x + 1.

e^0=1, so f(0) = 6+1 = 7. Its derivative is 6e^x, so f′(0) = 6.

L40-P026 | Compute f(0) and f′(0) for f(x) = 2e^x + 2.

e^0=1, so f(0) = 2+2 = 4. Its derivative is 2e^x, so f′(0) = 2.

L40-P027 | Compute f(0) and f′(0) for f(x) = 3e^x + 3.

e^0=1, so f(0) = 3+3 = 6. Its derivative is 3e^x, so f′(0) = 3.

L40-P028 | Compute f(0) and f′(0) for f(x) = 4e^x + 4.

e^0=1, so f(0) = 4+4 = 8. Its derivative is 4e^x, so f′(0) = 4.

L40-P029 | Compute f(0) and f′(0) for f(x) = 5e^x + 1.

e^0=1, so f(0) = 5+1 = 6. Its derivative is 5e^x, so f′(0) = 5.

L40-P030 | Compute f(0) and f′(0) for f(x) = 6e^x + 2.

e^0=1, so f(0) = 6+2 = 8. Its derivative is 6e^x, so f′(0) = 6.

L40-P031 | Compute f(0) and f′(0) for f(x) = 2e^x + 3.

e^0=1, so f(0) = 2+3 = 5. Its derivative is 2e^x, so f′(0) = 2.

L40-P032 | Compute f(0) and f′(0) for f(x) = 3e^x + 4.

e^0=1, so f(0) = 3+4 = 7. Its derivative is 3e^x, so f′(0) = 3.

L40-P033 | Compute f(0) and f′(0) for f(x) = 4e^x + 1.

e^0=1, so f(0) = 4+1 = 5. Its derivative is 4e^x, so f′(0) = 4.

L40-P034 | Compute f(0) and f′(0) for f(x) = 5e^x + 2.

e^0=1, so f(0) = 5+2 = 7. Its derivative is 5e^x, so f′(0) = 5.

L40-P035 | Compute f(0) and f′(0) for f(x) = 6e^x + 3.

e^0=1, so f(0) = 6+3 = 9. Its derivative is 6e^x, so f′(0) = 6.

L40-P036 | Compute f(0) and f′(0) for f(x) = 2e^x + 4.

e^0=1, so f(0) = 2+4 = 6. Its derivative is 2e^x, so f′(0) = 2.

L40-P037 | Compute f(0) and f′(0) for f(x) = 3e^x + 1.

e^0=1, so f(0) = 3+1 = 4. Its derivative is 3e^x, so f′(0) = 3.

L40-P038 | Compute f(0) and f′(0) for f(x) = 4e^x + 2.

e^0=1, so f(0) = 4+2 = 6. Its derivative is 4e^x, so f′(0) = 4.

L40-P039 | Compute f(0) and f′(0) for f(x) = 5e^x + 3.

e^0=1, so f(0) = 5+3 = 8. Its derivative is 5e^x, so f′(0) = 5.

L40-P040 | Compute f(0) and f′(0) for f(x) = 6e^x + 4.

e^0=1, so f(0) = 6+4 = 10. Its derivative is 6e^x, so f′(0) = 6.

3/5 | Challenger

L40-P041 | Differentiate f(x) = 2e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 2e^(1x + 1).

L40-P042 | Differentiate f(x) = 3e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 6e^(2x + 2).

L40-P043 | Differentiate f(x) = 4e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 12e^(3x + 3).

L40-P044 | Differentiate f(x) = 5e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 20e^(4x + 4).

L40-P045 | Differentiate f(x) = 6e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 6e^(1x + 1).

L40-P046 | Differentiate f(x) = 2e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 4e^(2x + 2).

L40-P047 | Differentiate f(x) = 3e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 9e^(3x + 3).

L40-P048 | Differentiate f(x) = 4e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 16e^(4x + 4).

L40-P049 | Differentiate f(x) = 5e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 5e^(1x + 1).

L40-P050 | Differentiate f(x) = 6e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 12e^(2x + 2).

L40-P051 | Differentiate f(x) = 2e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 6e^(3x + 3).

L40-P052 | Differentiate f(x) = 3e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 12e^(4x + 4).

L40-P053 | Differentiate f(x) = 4e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 4e^(1x + 1).

L40-P054 | Differentiate f(x) = 5e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 10e^(2x + 2).

L40-P055 | Differentiate f(x) = 6e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 18e^(3x + 3).

L40-P056 | Differentiate f(x) = 2e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 8e^(4x + 4).

L40-P057 | Differentiate f(x) = 3e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 3e^(1x + 1).

L40-P058 | Differentiate f(x) = 4e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 8e^(2x + 2).

L40-P059 | Differentiate f(x) = 5e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 15e^(3x + 3).

L40-P060 | Differentiate f(x) = 6e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 24e^(4x + 4).

4/5 | Expert

L40-P061 | Solve 2e^x = 2e^1 for real x.

Divide by nonzero 2: e^x=e^1; exponential is one-to-one, so x=1.

L40-P062 | Solve 3e^x = 3e^2 for real x.

Divide by nonzero 3: e^x=e^2; exponential is one-to-one, so x=2.

L40-P063 | Solve 4e^x = 4e^3 for real x.

Divide by nonzero 4: e^x=e^3; exponential is one-to-one, so x=3.

L40-P064 | Solve 5e^x = 5e^4 for real x.

Divide by nonzero 5: e^x=e^4; exponential is one-to-one, so x=4.

L40-P065 | Solve 6e^x = 6e^1 for real x.

Divide by nonzero 6: e^x=e^1; exponential is one-to-one, so x=1.

L40-P066 | Solve 2e^x = 2e^2 for real x.

Divide by nonzero 2: e^x=e^2; exponential is one-to-one, so x=2.

L40-P067 | Solve 3e^x = 3e^3 for real x.

Divide by nonzero 3: e^x=e^3; exponential is one-to-one, so x=3.

L40-P068 | Solve 4e^x = 4e^4 for real x.

Divide by nonzero 4: e^x=e^4; exponential is one-to-one, so x=4.

L40-P069 | Solve 5e^x = 5e^1 for real x.

Divide by nonzero 5: e^x=e^1; exponential is one-to-one, so x=1.

L40-P070 | Solve 6e^x = 6e^2 for real x.

Divide by nonzero 6: e^x=e^2; exponential is one-to-one, so x=2.

L40-P071 | Solve 2e^x = 2e^3 for real x.

Divide by nonzero 2: e^x=e^3; exponential is one-to-one, so x=3.

L40-P072 | Solve 3e^x = 3e^4 for real x.

Divide by nonzero 3: e^x=e^4; exponential is one-to-one, so x=4.

L40-P073 | Solve 4e^x = 4e^1 for real x.

Divide by nonzero 4: e^x=e^1; exponential is one-to-one, so x=1.

L40-P074 | Solve 5e^x = 5e^2 for real x.

Divide by nonzero 5: e^x=e^2; exponential is one-to-one, so x=2.

L40-P075 | Solve 6e^x = 6e^3 for real x.

Divide by nonzero 6: e^x=e^3; exponential is one-to-one, so x=3.

L40-P076 | Solve 2e^x = 2e^4 for real x.

Divide by nonzero 2: e^x=e^4; exponential is one-to-one, so x=4.

L40-P077 | Solve 3e^x = 3e^1 for real x.

Divide by nonzero 3: e^x=e^1; exponential is one-to-one, so x=1.

L40-P078 | Solve 4e^x = 4e^2 for real x.

Divide by nonzero 4: e^x=e^2; exponential is one-to-one, so x=2.

L40-P079 | Solve 5e^x = 5e^3 for real x.

Divide by nonzero 5: e^x=e^3; exponential is one-to-one, so x=3.

L40-P080 | Solve 6e^x = 6e^4 for real x.

Divide by nonzero 6: e^x=e^4; exponential is one-to-one, so x=4.

5/5 | Ninja

L40-P081 | A growth model is N(t)=2e^(1t); find N’(t), N’(0)/N(0), and interpret.

N’(t) = 2e^(1t). Ratio N’/N = 1 at every t: it is the instantaneous relative growth rate.

L40-P082 | A growth model is N(t)=3e^(2t); find N’(t), N’(1)/N(1), and interpret.

N’(t) = 6e^(2t). Ratio N’/N = 2 at every t: it is the instantaneous relative growth rate.

L40-P083 | A growth model is N(t)=4e^(3t); find N’(t), N’(2)/N(2), and interpret.

N’(t) = 12e^(3t). Ratio N’/N = 3 at every t: it is the instantaneous relative growth rate.

L40-P084 | A growth model is N(t)=5e^(4t); find N’(t), N’(3)/N(3), and interpret.

N’(t) = 20e^(4t). Ratio N’/N = 4 at every t: it is the instantaneous relative growth rate.

L40-P085 | A growth model is N(t)=6e^(1t); find N’(t), N’(0)/N(0), and interpret.

N’(t) = 6e^(1t). Ratio N’/N = 1 at every t: it is the instantaneous relative growth rate.

L40-P086 | A growth model is N(t)=2e^(2t); find N’(t), N’(1)/N(1), and interpret.

N’(t) = 4e^(2t). Ratio N’/N = 2 at every t: it is the instantaneous relative growth rate.

L40-P087 | A growth model is N(t)=3e^(3t); find N’(t), N’(2)/N(2), and interpret.

N’(t) = 9e^(3t). Ratio N’/N = 3 at every t: it is the instantaneous relative growth rate.

L40-P088 | A growth model is N(t)=4e^(4t); find N’(t), N’(3)/N(3), and interpret.

N’(t) = 16e^(4t). Ratio N’/N = 4 at every t: it is the instantaneous relative growth rate.

L40-P089 | A growth model is N(t)=5e^(1t); find N’(t), N’(0)/N(0), and interpret.

N’(t) = 5e^(1t). Ratio N’/N = 1 at every t: it is the instantaneous relative growth rate.

L40-P090 | A growth model is N(t)=6e^(2t); find N’(t), N’(1)/N(1), and interpret.

N’(t) = 12e^(2t). Ratio N’/N = 2 at every t: it is the instantaneous relative growth rate.

L40-P091 | A growth model is N(t)=2e^(3t); find N’(t), N’(2)/N(2), and interpret.

N’(t) = 6e^(3t). Ratio N’/N = 3 at every t: it is the instantaneous relative growth rate.

L40-P092 | A growth model is N(t)=3e^(4t); find N’(t), N’(3)/N(3), and interpret.

N’(t) = 12e^(4t). Ratio N’/N = 4 at every t: it is the instantaneous relative growth rate.

L40-P093 | A growth model is N(t)=4e^(1t); find N’(t), N’(0)/N(0), and interpret.

N’(t) = 4e^(1t). Ratio N’/N = 1 at every t: it is the instantaneous relative growth rate.

L40-P094 | A growth model is N(t)=5e^(2t); find N’(t), N’(1)/N(1), and interpret.

N’(t) = 10e^(2t). Ratio N’/N = 2 at every t: it is the instantaneous relative growth rate.

L40-P095 | A growth model is N(t)=6e^(3t); find N’(t), N’(2)/N(2), and interpret.

N’(t) = 18e^(3t). Ratio N’/N = 3 at every t: it is the instantaneous relative growth rate.

L40-P096 | A growth model is N(t)=2e^(4t); find N’(t), N’(3)/N(3), and interpret.

N’(t) = 8e^(4t). Ratio N’/N = 4 at every t: it is the instantaneous relative growth rate.

L40-P097 | A growth model is N(t)=3e^(1t); find N’(t), N’(0)/N(0), and interpret.

N’(t) = 3e^(1t). Ratio N’/N = 1 at every t: it is the instantaneous relative growth rate.

L40-P098 | A growth model is N(t)=4e^(2t); find N’(t), N’(1)/N(1), and interpret.

N’(t) = 8e^(2t). Ratio N’/N = 2 at every t: it is the instantaneous relative growth rate.

L40-P099 | A growth model is N(t)=5e^(3t); find N’(t), N’(2)/N(2), and interpret.

N’(t) = 15e^(3t). Ratio N’/N = 3 at every t: it is the instantaneous relative growth rate.

L40-P100 | A growth model is N(t)=6e^(4t); find N’(t), N’(3)/N(3), and interpret.

N’(t) = 24e^(4t). Ratio N’/N = 4 at every t: it is the instantaneous relative growth rate.

L41 | The natural logarithm

Source: PDF pages 16, 17

1/5 | Apprentice

L41-P001 | Compute ln(e^0) and e^(ln(2)).

Inverse functions: ln(e^0) = 0, and e^(ln(2)) = 2 (since 2>0).

L41-P002 | Compute ln(e^1) and e^(ln(3)).

Inverse functions: ln(e^1) = 1, and e^(ln(3)) = 3 (since 3>0).

L41-P003 | Compute ln(e^2) and e^(ln(4)).

Inverse functions: ln(e^2) = 2, and e^(ln(4)) = 4 (since 4>0).

L41-P004 | Compute ln(e^3) and e^(ln(5)).

Inverse functions: ln(e^3) = 3, and e^(ln(5)) = 5 (since 5>0).

L41-P005 | Compute ln(e^4) and e^(ln(6)).

Inverse functions: ln(e^4) = 4, and e^(ln(6)) = 6 (since 6>0).

L41-P006 | Compute ln(e^5) and e^(ln(2)).

Inverse functions: ln(e^5) = 5, and e^(ln(2)) = 2 (since 2>0).

L41-P007 | Compute ln(e^0) and e^(ln(3)).

Inverse functions: ln(e^0) = 0, and e^(ln(3)) = 3 (since 3>0).

L41-P008 | Compute ln(e^1) and e^(ln(4)).

Inverse functions: ln(e^1) = 1, and e^(ln(4)) = 4 (since 4>0).

L41-P009 | Compute ln(e^2) and e^(ln(5)).

Inverse functions: ln(e^2) = 2, and e^(ln(5)) = 5 (since 5>0).

L41-P010 | Compute ln(e^3) and e^(ln(6)).

Inverse functions: ln(e^3) = 3, and e^(ln(6)) = 6 (since 6>0).

L41-P011 | Compute ln(e^4) and e^(ln(2)).

Inverse functions: ln(e^4) = 4, and e^(ln(2)) = 2 (since 2>0).

L41-P012 | Compute ln(e^5) and e^(ln(3)).

Inverse functions: ln(e^5) = 5, and e^(ln(3)) = 3 (since 3>0).

L41-P013 | Compute ln(e^0) and e^(ln(4)).

Inverse functions: ln(e^0) = 0, and e^(ln(4)) = 4 (since 4>0).

L41-P014 | Compute ln(e^1) and e^(ln(5)).

Inverse functions: ln(e^1) = 1, and e^(ln(5)) = 5 (since 5>0).

L41-P015 | Compute ln(e^2) and e^(ln(6)).

Inverse functions: ln(e^2) = 2, and e^(ln(6)) = 6 (since 6>0).

L41-P016 | Compute ln(e^3) and e^(ln(2)).

Inverse functions: ln(e^3) = 3, and e^(ln(2)) = 2 (since 2>0).

L41-P017 | Compute ln(e^4) and e^(ln(3)).

Inverse functions: ln(e^4) = 4, and e^(ln(3)) = 3 (since 3>0).

L41-P018 | Compute ln(e^5) and e^(ln(4)).

Inverse functions: ln(e^5) = 5, and e^(ln(4)) = 4 (since 4>0).

L41-P019 | Compute ln(e^0) and e^(ln(5)).

Inverse functions: ln(e^0) = 0, and e^(ln(5)) = 5 (since 5>0).

L41-P020 | Compute ln(e^1) and e^(ln(6)).

Inverse functions: ln(e^1) = 1, and e^(ln(6)) = 6 (since 6>0).

2/5 | Practitioner

L41-P021 | Simplify ln(2) - ln(2); state the assumptions.

Both arguments are positive. ln(2) - ln(2) = ln((2)/2) = ln(1).

L41-P022 | Simplify ln(6) - ln(3); state the assumptions.

Both arguments are positive. ln(6) - ln(3) = ln((6)/3) = ln(2).

L41-P023 | Simplify ln(12) - ln(4); state the assumptions.

Both arguments are positive. ln(12) - ln(4) = ln((12)/4) = ln(3).

L41-P024 | Simplify ln(20) - ln(5); state the assumptions.

Both arguments are positive. ln(20) - ln(5) = ln((20)/5) = ln(4).

L41-P025 | Simplify ln(6) - ln(6); state the assumptions.

Both arguments are positive. ln(6) - ln(6) = ln((6)/6) = ln(1).

L41-P026 | Simplify ln(4) - ln(2); state the assumptions.

Both arguments are positive. ln(4) - ln(2) = ln((4)/2) = ln(2).

L41-P027 | Simplify ln(9) - ln(3); state the assumptions.

Both arguments are positive. ln(9) - ln(3) = ln((9)/3) = ln(3).

L41-P028 | Simplify ln(16) - ln(4); state the assumptions.

Both arguments are positive. ln(16) - ln(4) = ln((16)/4) = ln(4).

L41-P029 | Simplify ln(5) - ln(5); state the assumptions.

Both arguments are positive. ln(5) - ln(5) = ln((5)/5) = ln(1).

L41-P030 | Simplify ln(12) - ln(6); state the assumptions.

Both arguments are positive. ln(12) - ln(6) = ln((12)/6) = ln(2).

L41-P031 | Simplify ln(6) - ln(2); state the assumptions.

Both arguments are positive. ln(6) - ln(2) = ln((6)/2) = ln(3).

L41-P032 | Simplify ln(12) - ln(3); state the assumptions.

Both arguments are positive. ln(12) - ln(3) = ln((12)/3) = ln(4).

L41-P033 | Simplify ln(4) - ln(4); state the assumptions.

Both arguments are positive. ln(4) - ln(4) = ln((4)/4) = ln(1).

L41-P034 | Simplify ln(10) - ln(5); state the assumptions.

Both arguments are positive. ln(10) - ln(5) = ln((10)/5) = ln(2).

L41-P035 | Simplify ln(18) - ln(6); state the assumptions.

Both arguments are positive. ln(18) - ln(6) = ln((18)/6) = ln(3).

L41-P036 | Simplify ln(8) - ln(2); state the assumptions.

Both arguments are positive. ln(8) - ln(2) = ln((8)/2) = ln(4).

L41-P037 | Simplify ln(3) - ln(3); state the assumptions.

Both arguments are positive. ln(3) - ln(3) = ln((3)/3) = ln(1).

L41-P038 | Simplify ln(8) - ln(4); state the assumptions.

Both arguments are positive. ln(8) - ln(4) = ln((8)/4) = ln(2).

L41-P039 | Simplify ln(15) - ln(5); state the assumptions.

Both arguments are positive. ln(15) - ln(5) = ln((15)/5) = ln(3).

L41-P040 | Simplify ln(24) - ln(6); state the assumptions.

Both arguments are positive. ln(24) - ln(6) = ln((24)/6) = ln(4).

3/5 | Challenger

L41-P041 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P042 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P043 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P044 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

L41-P045 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P046 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P047 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P048 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

L41-P049 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P050 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P051 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P052 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

L41-P053 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P054 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P055 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P056 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

L41-P057 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P058 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P059 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P060 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

4/5 | Expert

L41-P061 | Differentiate f(x)=ln(2x + 1) and find f’(1); state the domain.

Require 2x + 1 > 0, so x > -1/2. Chain rule: f’(x)=2/(2x + 1); f’(1)=2/3.

L41-P062 | Differentiate f(x)=ln(3x + 2) and find f’(2); state the domain.

Require 3x + 2 > 0, so x > -2/3. Chain rule: f’(x)=3/(3x + 2); f’(2)=3/8.

L41-P063 | Differentiate f(x)=ln(4x + 3) and find f’(3); state the domain.

Require 4x + 3 > 0, so x > -3/4. Chain rule: f’(x)=4/(4x + 3); f’(3)=4/15.

L41-P064 | Differentiate f(x)=ln(5x + 4) and find f’(4); state the domain.

Require 5x + 4 > 0, so x > -4/5. Chain rule: f’(x)=5/(5x + 4); f’(4)=5/24.

L41-P065 | Differentiate f(x)=ln(6x + 1) and find f’(1); state the domain.

Require 6x + 1 > 0, so x > -1/6. Chain rule: f’(x)=6/(6x + 1); f’(1)=6/7.

L41-P066 | Differentiate f(x)=ln(2x + 2) and find f’(2); state the domain.

Require 2x + 2 > 0, so x > -2/2. Chain rule: f’(x)=2/(2x + 2); f’(2)=1/3.

L41-P067 | Differentiate f(x)=ln(3x + 3) and find f’(3); state the domain.

Require 3x + 3 > 0, so x > -3/3. Chain rule: f’(x)=3/(3x + 3); f’(3)=1/4.

L41-P068 | Differentiate f(x)=ln(4x + 4) and find f’(4); state the domain.

Require 4x + 4 > 0, so x > -4/4. Chain rule: f’(x)=4/(4x + 4); f’(4)=1/5.

L41-P069 | Differentiate f(x)=ln(5x + 1) and find f’(1); state the domain.

Require 5x + 1 > 0, so x > -1/5. Chain rule: f’(x)=5/(5x + 1); f’(1)=5/6.

L41-P070 | Differentiate f(x)=ln(6x + 2) and find f’(2); state the domain.

Require 6x + 2 > 0, so x > -2/6. Chain rule: f’(x)=6/(6x + 2); f’(2)=3/7.

L41-P071 | Differentiate f(x)=ln(2x + 3) and find f’(3); state the domain.

Require 2x + 3 > 0, so x > -3/2. Chain rule: f’(x)=2/(2x + 3); f’(3)=2/9.

L41-P072 | Differentiate f(x)=ln(3x + 4) and find f’(4); state the domain.

Require 3x + 4 > 0, so x > -4/3. Chain rule: f’(x)=3/(3x + 4); f’(4)=3/16.

L41-P073 | Differentiate f(x)=ln(4x + 1) and find f’(1); state the domain.

Require 4x + 1 > 0, so x > -1/4. Chain rule: f’(x)=4/(4x + 1); f’(1)=4/5.

L41-P074 | Differentiate f(x)=ln(5x + 2) and find f’(2); state the domain.

Require 5x + 2 > 0, so x > -2/5. Chain rule: f’(x)=5/(5x + 2); f’(2)=5/12.

L41-P075 | Differentiate f(x)=ln(6x + 3) and find f’(3); state the domain.

Require 6x + 3 > 0, so x > -3/6. Chain rule: f’(x)=6/(6x + 3); f’(3)=2/7.

L41-P076 | Differentiate f(x)=ln(2x + 4) and find f’(4); state the domain.

Require 2x + 4 > 0, so x > -4/2. Chain rule: f’(x)=2/(2x + 4); f’(4)=1/6.

L41-P077 | Differentiate f(x)=ln(3x + 1) and find f’(1); state the domain.

Require 3x + 1 > 0, so x > -1/3. Chain rule: f’(x)=3/(3x + 1); f’(1)=3/4.

L41-P078 | Differentiate f(x)=ln(4x + 2) and find f’(2); state the domain.

Require 4x + 2 > 0, so x > -2/4. Chain rule: f’(x)=4/(4x + 2); f’(2)=2/5.

L41-P079 | Differentiate f(x)=ln(5x + 3) and find f’(3); state the domain.

Require 5x + 3 > 0, so x > -3/5. Chain rule: f’(x)=5/(5x + 3); f’(3)=5/18.

L41-P080 | Differentiate f(x)=ln(6x + 4) and find f’(4); state the domain.

Require 6x + 4 > 0, so x > -4/6. Chain rule: f’(x)=6/(6x + 4); f’(4)=3/14.

5/5 | Ninja

L41-P081 | Let f(x)=ln((2x + 1)^2). State the real domain and find f’(1).

Argument > 0: 2x + 1 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=2×2/(2x + 1); f’(1)=4/3.

L41-P082 | Let f(x)=ln((3x + 2)^3). State the real domain and find f’(2).

Argument > 0: 3x + 2 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=3×3/(3x + 2); f’(2)=9/8.

L41-P083 | Let f(x)=ln((4x + 3)^4). State the real domain and find f’(3).

Argument > 0: 4x + 3 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=4×4/(4x + 3); f’(3)=16/15.

L41-P084 | Let f(x)=ln((5x + 4)^5). State the real domain and find f’(4).

Argument > 0: 5x + 4 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=5×5/(5x + 4); f’(4)=25/24.

L41-P085 | Let f(x)=ln((6x + 1)^6). State the real domain and find f’(1).

Argument > 0: 6x + 1 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=6×6/(6x + 1); f’(1)=36/7.

L41-P086 | Let f(x)=ln((2x + 2)^2). State the real domain and find f’(2).

Argument > 0: 2x + 2 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=2×2/(2x + 2); f’(2)=2/3.

L41-P087 | Let f(x)=ln((3x + 3)^3). State the real domain and find f’(3).

Argument > 0: 3x + 3 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=3×3/(3x + 3); f’(3)=3/4.

L41-P088 | Let f(x)=ln((4x + 4)^4). State the real domain and find f’(4).

Argument > 0: 4x + 4 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=4×4/(4x + 4); f’(4)=4/5.

L41-P089 | Let f(x)=ln((5x + 1)^5). State the real domain and find f’(1).

Argument > 0: 5x + 1 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=5×5/(5x + 1); f’(1)=25/6.

L41-P090 | Let f(x)=ln((6x + 2)^6). State the real domain and find f’(2).

Argument > 0: 6x + 2 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=6×6/(6x + 2); f’(2)=18/7.

L41-P091 | Let f(x)=ln((2x + 3)^2). State the real domain and find f’(3).

Argument > 0: 2x + 3 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=2×2/(2x + 3); f’(3)=4/9.

L41-P092 | Let f(x)=ln((3x + 4)^3). State the real domain and find f’(4).

Argument > 0: 3x + 4 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=3×3/(3x + 4); f’(4)=9/16.

L41-P093 | Let f(x)=ln((4x + 1)^4). State the real domain and find f’(1).

Argument > 0: 4x + 1 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=4×4/(4x + 1); f’(1)=16/5.

L41-P094 | Let f(x)=ln((5x + 2)^5). State the real domain and find f’(2).

Argument > 0: 5x + 2 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=5×5/(5x + 2); f’(2)=25/12.

L41-P095 | Let f(x)=ln((6x + 3)^6). State the real domain and find f’(3).

Argument > 0: 6x + 3 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=6×6/(6x + 3); f’(3)=12/7.

L41-P096 | Let f(x)=ln((2x + 4)^2). State the real domain and find f’(4).

Argument > 0: 2x + 4 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=2×2/(2x + 4); f’(4)=1/3.

L41-P097 | Let f(x)=ln((3x + 1)^3). State the real domain and find f’(1).

Argument > 0: 3x + 1 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=3×3/(3x + 1); f’(1)=9/4.

L41-P098 | Let f(x)=ln((4x + 2)^4). State the real domain and find f’(2).

Argument > 0: 4x + 2 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=4×4/(4x + 2); f’(2)=8/5.

L41-P099 | Let f(x)=ln((5x + 3)^5). State the real domain and find f’(3).

Argument > 0: 5x + 3 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=5×5/(5x + 3); f’(3)=25/18.

L41-P100 | Let f(x)=ln((6x + 4)^6). State the real domain and find f’(4).

Argument > 0: 6x + 4 ≠ 0 for even n, > 0 for odd n. On its domain f’(x)=6×6/(6x + 4); f’(4)=9/7.

L42 | Bases, logarithms and complexity

Source: PDF pages 18, 19

1/5 | Apprentice

L42-P001 | Compute log_2(2^1) and 2^(log_2(2)).

Base-2 logarithms and exponentials are inverse: 1, then 2.

L42-P002 | Compute log_3(3^2) and 3^(log_3(9)).

Base-3 logarithms and exponentials are inverse: 2, then 9.

L42-P003 | Compute log_4(4^3) and 4^(log_4(64)).

Base-4 logarithms and exponentials are inverse: 3, then 64.

L42-P004 | Compute log_5(5^4) and 5^(log_5(625)).

Base-5 logarithms and exponentials are inverse: 4, then 625.

L42-P005 | Compute log_2(2^5) and 2^(log_2(32)).

Base-2 logarithms and exponentials are inverse: 5, then 32.

L42-P006 | Compute log_3(3^6) and 3^(log_3(729)).

Base-3 logarithms and exponentials are inverse: 6, then 729.

L42-P007 | Compute log_4(4^1) and 4^(log_4(4)).

Base-4 logarithms and exponentials are inverse: 1, then 4.

L42-P008 | Compute log_5(5^2) and 5^(log_5(25)).

Base-5 logarithms and exponentials are inverse: 2, then 25.

L42-P009 | Compute log_2(2^3) and 2^(log_2(8)).

Base-2 logarithms and exponentials are inverse: 3, then 8.

L42-P010 | Compute log_3(3^4) and 3^(log_3(81)).

Base-3 logarithms and exponentials are inverse: 4, then 81.

L42-P011 | Compute log_4(4^5) and 4^(log_4(1024)).

Base-4 logarithms and exponentials are inverse: 5, then 1024.

L42-P012 | Compute log_5(5^6) and 5^(log_5(15625)).

Base-5 logarithms and exponentials are inverse: 6, then 15625.

L42-P013 | Compute log_2(2^1) and 2^(log_2(2)).

Base-2 logarithms and exponentials are inverse: 1, then 2.

L42-P014 | Compute log_3(3^2) and 3^(log_3(9)).

Base-3 logarithms and exponentials are inverse: 2, then 9.

L42-P015 | Compute log_4(4^3) and 4^(log_4(64)).

Base-4 logarithms and exponentials are inverse: 3, then 64.

L42-P016 | Compute log_5(5^4) and 5^(log_5(625)).

Base-5 logarithms and exponentials are inverse: 4, then 625.

L42-P017 | Compute log_2(2^5) and 2^(log_2(32)).

Base-2 logarithms and exponentials are inverse: 5, then 32.

L42-P018 | Compute log_3(3^6) and 3^(log_3(729)).

Base-3 logarithms and exponentials are inverse: 6, then 729.

L42-P019 | Compute log_4(4^1) and 4^(log_4(4)).

Base-4 logarithms and exponentials are inverse: 1, then 4.

L42-P020 | Compute log_5(5^2) and 5^(log_5(25)).

Base-5 logarithms and exponentials are inverse: 2, then 25.

2/5 | Practitioner

L42-P021 | Differentiate f(x)=2^x; compute f’(0).

Write 2^x=e^(x ln 2); hence f’(x)=ln(2)×2^x. At x=0: f’(0)=ln(2).

L42-P022 | Differentiate f(x)=3^x; compute f’(0).

Write 3^x=e^(x ln 3); hence f’(x)=ln(3)×3^x. At x=0: f’(0)=ln(3).

L42-P023 | Differentiate f(x)=4^x; compute f’(0).

Write 4^x=e^(x ln 4); hence f’(x)=ln(4)×4^x. At x=0: f’(0)=ln(4).

L42-P024 | Differentiate f(x)=5^x; compute f’(0).

Write 5^x=e^(x ln 5); hence f’(x)=ln(5)×5^x. At x=0: f’(0)=ln(5).

L42-P025 | Differentiate f(x)=2^x; compute f’(0).

Write 2^x=e^(x ln 2); hence f’(x)=ln(2)×2^x. At x=0: f’(0)=ln(2).

L42-P026 | Differentiate f(x)=3^x; compute f’(0).

Write 3^x=e^(x ln 3); hence f’(x)=ln(3)×3^x. At x=0: f’(0)=ln(3).

L42-P027 | Differentiate f(x)=4^x; compute f’(0).

Write 4^x=e^(x ln 4); hence f’(x)=ln(4)×4^x. At x=0: f’(0)=ln(4).

L42-P028 | Differentiate f(x)=5^x; compute f’(0).

Write 5^x=e^(x ln 5); hence f’(x)=ln(5)×5^x. At x=0: f’(0)=ln(5).

L42-P029 | Differentiate f(x)=2^x; compute f’(0).

Write 2^x=e^(x ln 2); hence f’(x)=ln(2)×2^x. At x=0: f’(0)=ln(2).

L42-P030 | Differentiate f(x)=3^x; compute f’(0).

Write 3^x=e^(x ln 3); hence f’(x)=ln(3)×3^x. At x=0: f’(0)=ln(3).

L42-P031 | Differentiate f(x)=4^x; compute f’(0).

Write 4^x=e^(x ln 4); hence f’(x)=ln(4)×4^x. At x=0: f’(0)=ln(4).

L42-P032 | Differentiate f(x)=5^x; compute f′(0).

Write 5^x=e^(x ln 5); hence f′(x)=ln(5)×5^x. At x=0: f′(0)=ln(5).

L42-P033 | Differentiate f(x)=2^x; compute f′(0).

Write 2^x=e^(x ln 2); hence f′(x)=ln(2)×2^x. At x=0: f′(0)=ln(2).

L42-P034 | Differentiate f(x)=3^x; compute f′(0).

Write 3^x=e^(x ln 3); hence f′(x)=ln(3)×3^x. At x=0: f′(0)=ln(3).

L42-P035 | Differentiate f(x)=4^x; compute f′(0).

Write 4^x=e^(x ln 4); hence f′(x)=ln(4)×4^x. At x=0: f′(0)=ln(4).

L42-P036 | Differentiate f(x)=5^x; compute f′(0).

Write 5^x=e^(x ln 5); hence f′(x)=ln(5)×5^x. At x=0: f′(0)=ln(5).

L42-P037 | Differentiate f(x)=2^x; compute f′(0).

Write 2^x=e^(x ln 2); hence f′(x)=ln(2)×2^x. At x=0: f′(0)=ln(2).

L42-P038 | Differentiate f(x)=3^x; compute f′(0).

Write 3^x=e^(x ln 3); hence f′(x)=ln(3)×3^x. At x=0: f′(0)=ln(3).

L42-P039 | Differentiate f(x)=4^x; compute f′(0).

Write 4^x=e^(x ln 4); hence f′(x)=ln(4)×4^x. At x=0: f′(0)=ln(4).

L42-P040 | Differentiate f(x)=5^x; compute f′(0).

Write 5^x=e^(x ln 5); hence f′(x)=ln(5)×5^x. At x=0: f′(0)=ln(5).

3/5 | Challenger

L42-P041 | Find log_2(2) using change of base.

log_2(2)=ln(2)/ln(2)=1ln(2)/ln(2)=1.

L42-P042 | Find log_3(9) using change of base.

log_3(9)=ln(9)/ln(3)=2ln(3)/ln(3)=2.

L42-P043 | Find log_4(64) using change of base.

log_4(64)=ln(64)/ln(4)=3ln(4)/ln(4)=3.

L42-P044 | Find log_5(625) using change of base.

log_5(625)=ln(625)/ln(5)=4ln(5)/ln(5)=4.

L42-P045 | Find log_2(32) using change of base.

log_2(32)=ln(32)/ln(2)=5ln(2)/ln(2)=5.

L42-P046 | Find log_3(729) using change of base.

log_3(729)=ln(729)/ln(3)=6ln(3)/ln(3)=6.

L42-P047 | Find log_4(4) using change of base.

log_4(4)=ln(4)/ln(4)=1ln(4)/ln(4)=1.

L42-P048 | Find log_5(25) using change of base.

log_5(25)=ln(25)/ln(5)=2ln(5)/ln(5)=2.

L42-P049 | Find log_2(8) using change of base.

log_2(8)=ln(8)/ln(2)=3ln(2)/ln(2)=3.

L42-P050 | Find log_3(81) using change of base.

log_3(81)=ln(81)/ln(3)=4ln(3)/ln(3)=4.

L42-P051 | Find log_4(1024) using change of base.

log_4(1024)=ln(1024)/ln(4)=5ln(4)/ln(4)=5.

L42-P052 | Find log_5(15625) using change of base.

log_5(15625)=ln(15625)/ln(5)=6ln(5)/ln(5)=6.

L42-P053 | Find log_2(2) using change of base.

log_2(2)=ln(2)/ln(2)=1ln(2)/ln(2)=1.

L42-P054 | Find log_3(9) using change of base.

log_3(9)=ln(9)/ln(3)=2ln(3)/ln(3)=2.

L42-P055 | Find log_4(64) using change of base.

log_4(64)=ln(64)/ln(4)=3ln(4)/ln(4)=3.

L42-P056 | Find log_5(625) using change of base.

log_5(625)=ln(625)/ln(5)=4ln(5)/ln(5)=4.

L42-P057 | Find log_2(32) using change of base.

log_2(32)=ln(32)/ln(2)=5ln(2)/ln(2)=5.

L42-P058 | Find log_3(729) using change of base.

log_3(729)=ln(729)/ln(3)=6ln(3)/ln(3)=6.

L42-P059 | Find log_4(4) using change of base.

log_4(4)=ln(4)/ln(4)=1ln(4)/ln(4)=1.

L42-P060 | Find log_5(25) using change of base.

log_5(25)=ln(25)/ln(5)=2ln(5)/ln(5)=2.

4/5 | Expert

L42-P061 | Differentiate f(x)=log_2(2x + 2) and find f’(1); state the domain.

Domain x > -2/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 2) ln(2)]; at 1: 1/2/ln(2).

L42-P062 | Differentiate f(x)=log_3(3x + 3) and find f’(2); state the domain.

Domain x > -3/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 3) ln(3)]; at 2: 1/3/ln(3).

L42-P063 | Differentiate f(x)=log_4(4x + 4) and find f’(3); state the domain.

Domain x > -4/4. Since log_4(u)=ln(u)/ln(4), f’(x)=4/[(4x + 4) ln(4)]; at 3: 1/4/ln(4).

L42-P064 | Differentiate f(x)=log_5(5x + 2) and find f’(4); state the domain.

Domain x > -2/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 2) ln(5)]; at 4: 5/22/ln(5).

L42-P065 | Differentiate f(x)=log_2(2x + 3) and find f’(1); state the domain.

Domain x > -3/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 3) ln(2)]; at 1: 2/5/ln(2).

L42-P066 | Differentiate f(x)=log_3(3x + 4) and find f’(2); state the domain.

Domain x > -4/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 4) ln(3)]; at 2: 3/10/ln(3).

L42-P067 | Differentiate f(x)=log_4(4x + 2) and find f’(3); state the domain.

Domain x > -2/4. Since log_4(u)=ln(u)/ln(4), f’(x)=4/[(4x + 2) ln(4)]; at 3: 2/7/ln(4).

L42-P068 | Differentiate f(x)=log_5(5x + 3) and find f’(4); state the domain.

Domain x > -3/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 3) ln(5)]; at 4: 5/23/ln(5).

L42-P069 | Differentiate f(x)=log_2(2x + 4) and find f’(1); state the domain.

Domain x > -4/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 4) ln(2)]; at 1: 1/3/ln(2).

L42-P070 | Differentiate f(x)=log_3(3x + 2) and find f’(2); state the domain.

Domain x > -2/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 2) ln(3)]; at 2: 3/8/ln(3).

L42-P071 | Differentiate f(x)=log_4(4x + 3) and find f’(3); state the domain.

Domain x > -3/4. Since log_4(u)=ln(u)/ln(4), f’(x)=4/[(4x + 3) ln(4)]; at 3: 4/15/ln(4).

L42-P072 | Differentiate f(x)=log_5(5x + 4) and find f’(4); state the domain.

Domain x > -4/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 4) ln(5)]; at 4: 5/24/ln(5).

L42-P073 | Differentiate f(x)=log_2(2x + 2) and find f’(1); state the domain.

Domain x > -2/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 2) ln(2)]; at 1: 1/2/ln(2).

L42-P074 | Differentiate f(x)=log_3(3x + 3) and find f’(2); state the domain.

Domain x > -3/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 3) ln(3)]; at 2: 1/3/ln(3).

L42-P075 | Differentiate f(x)=log_4(4x + 4) and find f’(3); state the domain.

Domain x > -4/4. Since log_4(u)=ln(u)/ln(4), f’(x)=4/[(4x + 4) ln(4)]; at 3: 1/4/ln(4).

L42-P076 | Differentiate f(x)=log_5(5x + 2) and find f’(4); state the domain.

Domain x > -2/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 2) ln(5)]; at 4: 5/22/ln(5).

L42-P077 | Differentiate f(x)=log_2(2x + 3) and find f’(1); state the domain.

Domain x > -3/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 3) ln(2)]; at 1: 2/5/ln(2).

L42-P078 | Differentiate f(x)=log_3(3x + 4) and find f’(2); state the domain.

Domain x > -4/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 4) ln(3)]; at 2: 3/10/ln(3).

L42-P079 | Differentiate f(x)=log_4(4x + 2) and find f’(3); state the domain.

Domain x > -2/4. Since log_4(u)=ln(u)/ln(4), f’(x)=4/[(4x + 2) ln(4)]; at 3: 2/7/ln(4).

L42-P080 | Differentiate f(x)=log_5(5x + 3) and find f’(4); state the domain.

Domain x > -3/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 3) ln(5)]; at 4: 5/23/ln(5).

5/5 | Ninja

L42-P081 | At n=2^3=8, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(8)=3$. Thus n $\log_2(n)=8\times3=24$ and n^2=64. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P082 | At n=2^4=16, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(16)=4$. Thus n $\log_2(n)=16\times4=64$ and n^2=256. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P083 | At n=2^5=32, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(32)=5$. Thus n $\log_2(n)=32\times5=160$ and n^2=1024. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P084 | At n=2^6=64, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(64)=6$. Thus n $\log_2(n)=64\times6=384$ and n^2=4096. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P085 | At n=2^7=128, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(128)=7$. Thus n $\log_2(n)=128\times7=896$ and n^2=16384. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P086 | At n=2^8=256, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(256)=8$. Thus n $\log_2(n)=256\times8=2048$ and n^2=65536. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P087 | At n=2^9=512, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(512)=9$. Thus n $\log_2(n)=512\times9=4608$ and n^2=262144. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P088 | At n=2^3=8, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(8)=3$. Thus n $\log_2(n)=8\times3=24$ and n^2=64. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P089 | At n=2^4=16, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(16)=4$. Thus n $\log_2(n)=16\times4=64$ and n^2=256. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P090 | At n=2^5=32, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(32)=5$. Thus n $\log_2(n)=32\times5=160$ and n^2=1024. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P091 | At n=2^6=64, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(64)=6$. Thus n $\log_2(n)=64\times6=384$ and n^2=4096. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P092 | At n=2^7=128, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(128)=7$. Thus n $\log_2(n)=128\times7=896$ and n^2=16384. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P093 | At n=2^8=256, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(256)=8$. Thus n $\log_2(n)=256\times8=2048$ and n^2=65536. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P094 | At n=2^9=512, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(512)=9$. Thus n $\log_2(n)=512\times9=4608$ and n^2=262144. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P095 | At n=2^3=8, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(8)=3$. Thus n $\log_2(n)=8\times3=24$ and n^2=64. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P096 | At n=2^4=16, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(16)=4$. Thus n $\log_2(n)=16\times4=64$ and n^2=256. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P097 | At n=2^5=32, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(32)=5$. Thus n $\log_2(n)=32\times5=160$ and n^2=1024. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P098 | At n=2^6=64, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(64)=6$. Thus n $\log_2(n)=64\times6=384$ and n^2=4096. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P099 | At n=2^7=128, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(128)=7$. Thus n $\log_2(n)=128\times7=896$ and n^2=16384. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.

L42-P100 | At n=2^8=256, compare n $\log_2(n)$ with n^2. Which algorithm uses fewer operations at this n?

$\log_2(256)=8$. Thus n $\log_2(n)=256\times8=2048$ and n^2=65536. Here n $\log_2$ n is smaller. As n grows, its asymptotic growth is slower than n^2.